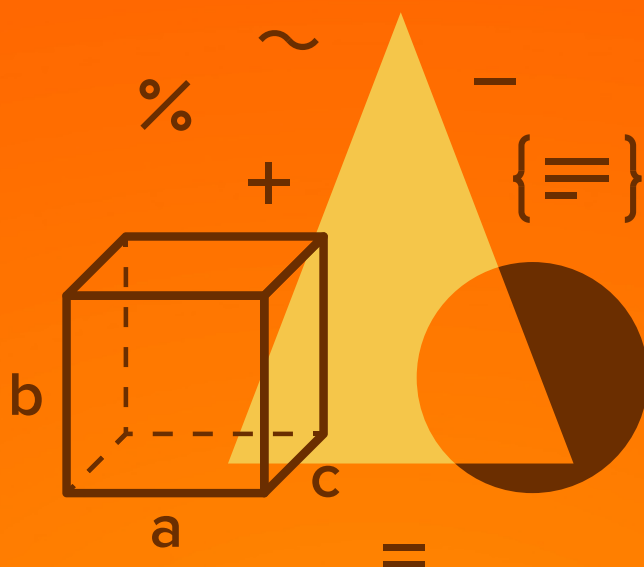




MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Indefinite & Definite Integration

English Medium

EXERCISE (O-1)

Straight Objective Type

1. Let $g(x)$ be an antiderivative for $f(x)$. Then $\ln(1 + (g(x))^2)$ is an antiderivative for
- (A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) none
- II0001
2. A function $y = f(x)$ satisfies $f''(x) = -\frac{1}{x^2} - \pi^2 \sin(\pi x)$; $f'(2) = \pi + \frac{1}{2}$ and $f(1) = 0$. The value of $f\left(\frac{1}{2}\right)$ is
- (A) $\ln 2$ (B) 1 (C) $\frac{\pi}{2} - \ln 2$ (D) $1 - \ln 2$
- II0002
3. Consider $f(x) = \frac{x^2}{1+x^3}$; $g(t) = \int f(t)dt$. If $g(1) = 0$ then $g(x)$ equals -
- (A) $\frac{1}{3} \ln(1+x^3)$ (B) $\frac{1}{3} \ln\left(\frac{1+x^3}{2}\right)$ (C) $\frac{1}{2} \ln\left(\frac{1+x^3}{3}\right)$ (D) $\frac{1}{3} \ln\left(\frac{1+x^3}{3}\right)$
- II0003
4. $\int \frac{x^2(1-\ln x)}{\ln^4 x - x^4} dx$ equals
- (A) $\frac{1}{2} \ln \left| \frac{x}{\ln x} \right| - \frac{1}{4} \ln |\ln^2 x - x^2| + C$ (B) $\frac{1}{4} \ln \left| \frac{\ln x - x}{\ln x + x} \right| - \frac{1}{2} \tan^{-1} \left(\frac{\ln x}{x} \right) + C$
- (C) $\frac{1}{4} \ln \left| \frac{\ln x + x}{\ln x - x} \right| + \frac{1}{2} \tan^{-1} \left(\frac{\ln x}{x} \right) + C$ (D) $\frac{1}{4} \left(\ln \left| \frac{\ln x - x}{\ln x + x} \right| + \tan^{-1} \left(\frac{\ln x}{x} \right) \right) + C$
- II0004
5. If $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$, where $f(x)$ is of the form of $ax^2 + bx + c$ then $(a+b+c)$ equals
- (A) 4 (B) 5 (C) 6 (D) none
- II0005

6. $\int \frac{x^3}{(2x^2 + 1)^3} dx$ is equal to -

(A) $\frac{1}{4} \left(2 + \frac{1}{x^2} \right)^{-2} + C$ (B) $-\frac{1}{4} \left(2 + \frac{1}{x^2} \right)^{-2} + C$ (C) $\frac{1}{2} \left(2 + \frac{1}{x^2} \right)^{-2} + C$ (D) $\frac{1}{4} \left(2 + \frac{1}{x^2} \right)^2 + C$

(where 'C' is integration constant)

II0006

7. Primitive of $\frac{3x^4 - 1}{(x^4 + x + 1)^2}$ w.r.t. x is -

(A) $\frac{x}{x^4 + x + 1} + c$ (B) $-\frac{x}{x^4 + x + 1} + c$ (C) $\frac{x+1}{x^4 + x + 1} + c$ (D) $-\frac{x+1}{x^4 + x + 1} + c$

II0007

8. $\int \frac{x^2 + x}{(e^x + x + 1)^2} dx$ equals

(A) $C - \ln(1 + (x+1)e^x) - \frac{1}{1 + (x+1)e^x}$ (B) $C - \ln(1 + (x+1)e^x) + \frac{1}{1 + (x-1)e^x}$
(C) $C - \ln(1 - (x^2 - 1)e^{-x}) + \frac{1}{1 + (x+1)e^x}$ (D) $C - \ln(1 + (x+1)e^{-x}) - \frac{1}{1 + (x+1)e^{-x}}$

II0056

9. $\int \frac{dx}{\sqrt[3]{x^{5/2}(x+1)^{7/2}}}$

(A) $-\left(\frac{x+1}{x}\right)^{1/6} + C$ (B) $6\left(\frac{x+1}{x}\right)^{-1/6} + C$ (C) $\left(\frac{x}{x+1}\right)^{5/6} + C$ (D) $-\left(\frac{x}{x+1}\right)^{5/6} + C$

II0009

10. $\int \frac{1-x^7}{x(1+x^7)} dx$ equals -

(A) $\ln x + \frac{2}{7} \ln(1+x^7) + c$ (B) $\ln x - \frac{2}{7} \ln(1-x^7) + c$
(C) $\ln x - \frac{2}{7} \ln(1+x^7) + c$ (D) $\ln x + \frac{2}{7} \ln(1-x^7) + c$

II0010

11. Integral of $\sqrt{1+2\cot x(\cot x + \operatorname{cosec} x)}$ w.r.t. x is

(A) $2 \ln \cos \frac{x}{2} + c$

(B) $2 \ln \sin \frac{x}{2} + c$

(C) $\frac{1}{2} \ln \cos \frac{x}{2} + c$

(D) $\ln \sin x - \ln(\operatorname{cosec} x - \cot x) + c$

II0011

12. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$

(A) $2e^{\sqrt{x}} [x - \sqrt{x} + 1] + C$

(B) $e^{\sqrt{x}} [x - 2\sqrt{x} + 1]$

(C) $e^{\sqrt{x}} (x + \sqrt{x}) + C$

(D) $e^{\sqrt{x}} (x + \sqrt{x} + 1) + C$

II0012

13. Let $f(x) = \frac{2\sin^2 x - 1}{\cos x} + \frac{\cos x(2\sin x + 1)}{1 + \sin x}$ then $\int e^x (f(x) + f'(x)) dx$ where c is the constant of integration)

(A) $e^x \tan x + c$

(B) $e^x \cot x + c$

(C) $e^x \operatorname{cosec}^2 x + c$

(D) $e^x \sec^2 x + c$

II0013

14. $\int e^x \left(\frac{x^2 - 3}{(x-1)^2} \right) dx$ is equal to -

(A) $e^x \left(\frac{x+3}{(x-1)} \right) + C$

(B) $e^x \left(\frac{x-3}{(x-1)} \right) + C$

(C) $e^x \left(\frac{x+1}{(x-1)} \right) + C$

(D) $e^x \left(\frac{1}{(x-1)} \right)^2 + C$

(where C is constant of integration)

II0014

15. $\int x \cdot \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ equals -

(A) $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) - x + c$

(B) $\frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + c$

(C) $\frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + c$

(D) $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) + x + c$

II0015

16. The evaluation of $\int \frac{p x^{p+2q-1} - q x^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$ is

- (A) $-\frac{x^p}{x^{p+q}+1} + C$ (B) $\frac{x^q}{x^{p+q}+1} + C$ (C) $-\frac{x^q}{x^{p+q}+1} + C$ (D) $\frac{x^p}{x^{p+q}+1} + C$

II0017

17. The integral $\int \sqrt{\cot x} e^{\sqrt{\sin x}} \sqrt{\cos x} dx$ equals

- (A) $\frac{\sqrt{\tan x} e^{\sqrt{\sin x}}}{\sqrt{\cos x}} + C$ (B) $2e^{\sqrt{\sin x}} + C$ (C) $-\frac{1}{2}e^{\sqrt{\sin x}} + C$ (D) $\frac{\sqrt{\cot x} e^{\sqrt{\sin x}}}{2\sqrt{\cos x}} + C$

II0018

18. $\int \frac{\cot^{-1}(e^x)}{e^x} dx$ is equal to -

- (A) $\frac{1}{2} \ln(e^{2x} + 1) - \frac{\cot^{-1}(e^x)}{e^x} + x + c$ (B) $\frac{1}{2} \ln(e^{2x} + 1) + \frac{\cot^{-1}(e^x)}{e^x} + x + c$
(C) $\frac{1}{2} \ln(e^{2x} + 1) - \frac{\cot^{-1}(e^x)}{e^x} - x + c$ (D) $\frac{1}{2} \ln(e^{2x} + 1) + \frac{\cot^{-1}(e^x)}{e^x} - x + c$

II0019

19. $\int (\sin(101x) \cdot \sin^{99} x) dx$ equals

- (A) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$ (B) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$
(C) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$ (D) $\frac{\sin(100x)(\sin x)^{101}}{101} + C$

II0020

20. $\int \sec^2 \theta (\sec \theta + \tan \theta)^2 d\theta$

- (A) $\frac{(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$ (B) $\frac{(\sec \theta + \tan \theta)}{3} [2 + 4 \tan \theta (\sec \theta + \tan \theta)] + C$
(C) $\frac{(\sec \theta + \tan \theta)}{3} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$ (D) $\frac{3(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

II0021

21. $\int \left[\sin \alpha \sin(x - \alpha) + \sin^2 \left(\frac{x}{2} - \alpha \right) \right] dx$ equals

(A) $\frac{1}{2}(x + \sin x) + C$

(B) $\frac{1}{2}(x^2 - \sin x) + C$

(C) $\frac{1}{2}(x - \sin x) + C$

(D) $\frac{1}{2}(x - \cos x) + C$

II0032

22. $\int \frac{dx}{(x - \alpha)\sqrt{(x - \alpha)(x - \beta)}}$ equals

(A) $\frac{2}{\alpha + \beta} \sqrt{\frac{x + \beta}{x + \alpha}} + C$

(B) $\frac{-2}{\alpha - \beta} \sqrt{\frac{x - \beta}{x - \alpha}} + C$

(C) $\frac{-2}{\alpha - \beta} \sqrt{\frac{x + \beta}{x + \alpha}} + C$

(D) $\frac{2}{\alpha + \beta} \sqrt{\frac{x - \beta}{x + \alpha}} + C$

II0035

23. $\int \frac{(ax^2 - b) dx}{x\sqrt{c^2x^2 - (ax^2 + b)^2}}$ equals

(A) $\sin^{-1} \left(\frac{ax^2 + b}{cx} \right) + k$

(B) $\sin^{-1} \left(\frac{ax^2 - b}{cx} \right) + k$

(C) $\sin^{-1} \left(\frac{a + bx^2}{cx} \right) + k$

(D) $\sin^{-1} \left(\frac{ax + b}{c} \right) + k$

II0038

24. $\int \left[\frac{\sqrt{x^2 + 1} [\ell n(x^2 + 1) - 2\ell n x]}{x^4} \right] dx$ equals

(A) $\frac{(x^2 + 1)\sqrt{x + 1}}{9x^2} \left[2 - 3\ell n \left(1 + \frac{1}{x^2} \right) \right]$

(B) $\frac{(x^2 - 1)\sqrt{x^2 - 1}}{9x} \left[2 - 3\ell n \left(1 + \frac{1}{x^2} \right) \right]$

(C) $\frac{(1 - x^2)\sqrt{1 - x^2}}{9} \left[2 - 3\ell n \left(1 + \frac{1}{x^2} \right) \right]$

(D) $\frac{(x^2 + 1)\sqrt{x^2 + 1}}{9x^3} \left[2 - 3\ell n \left(1 + \frac{1}{x^2} \right) \right]$

II0039

25. $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$ equals

(A) $(a+x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$

(B) $(a+x) \arctan \sqrt{\frac{a}{x^2}} - \sqrt{ax} + C$

(C) $(a+x^2) \arctan \sqrt{\frac{x^3}{a}} + \sqrt{ax} + C$

(D) $(a+x) \arctan \sqrt{x} + \sqrt{\frac{a}{x}} + C$

II0045

26. $\int \frac{dx}{\sin x + \sec x}$ equals

(A) $\frac{1}{\sqrt{3}} \ln \frac{\sqrt{3} + \sin x + \cos x}{\sqrt{3} - \sin x - \cos x} + \arctan(\sin x - \cos x) + C$

(B) $\frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} + \arctan(\sin x + \cos x) + C$

(C) $\frac{1}{2} \ln \frac{2 + \sin x - \cos x}{2 - \sin x + \cos x} + \arctan \sqrt{2}(\sin x - \cos x) + C$

(D) $\frac{1}{4\sqrt{3}} \ln \frac{\sqrt{2} + \sin x - \cos x}{\sqrt{2} - \sin x + \cos x} + \arctan \sqrt{3}(\sin x + \cos x) + C$

II0049

27. $\int \frac{4x^5 - 7x^4 + 8x^3 - 2x^2 + 4x - 7}{x^2(x^2 + 1)^2} dx$ equals

(A) $4 \ln x - \frac{7}{x} - 6 \tan^{-1}(x) - \frac{6x}{1-x^2} + C$

(B) $2 \ln x + \frac{1}{x} + 6 \tan^{-1}(x^2) + \frac{6x}{1-x^2} + C$

(C) $\ln x - \frac{1}{x} + \tan^{-1}(x) + \frac{x}{1-x^2} + C$

(D) $4 \ln x + \frac{7}{x} + 6 \tan^{-1}(x) + \frac{6x}{1+x^2} + C$

II0052

28. $\int \frac{(e^{\sqrt{x}} - e^{-\sqrt{x}}) \cos\left(e^{\sqrt{x}} + e^{-\sqrt{x}} + \frac{\pi}{4}\right) + (e^{\sqrt{x}} + e^{-\sqrt{x}}) \cos\left(e^{\sqrt{x}} - e^{-\sqrt{x}} + \frac{\pi}{4}\right)}{\sqrt{x}} dx$ equals

(A) $2\sqrt{2} \left(\cos(e^{-\sqrt{x}}) \right) \left(\sin(e^{\sqrt{x}}) + \cos(e^{\sqrt{x}}) \right) + C$

(B) $\left(\cos(e^{\sqrt{x}}) \right) \left(\sin(e^{-\sqrt{x}}) + \cos(e^{\sqrt{x}}) \right) + C$

(C) $\sqrt{2} \left(\cos(e^{-\sqrt{x}}) \right) \left(\sin(e^{-\sqrt{x}}) + \cos(e^{-\sqrt{x}}) \right) + C$

(D) $2 \left(\sin(e^{\sqrt{x}}) \right) \left(\sin(e^{-\sqrt{x}}) + \cos(e^{\sqrt{x}}) \right) + C$

II0053

29. $\int \frac{3x^2 + 1}{(x^2 - 1)^3} dx$ equals (where K is constant of integration)

- (A) $\frac{x}{(x^2 - 1)^2} + K$ (B) $K - \frac{x}{(x^2 + 1)^2}$ (C) $K - \frac{x}{(x^2 - 1)^2}$ (D) $K - \frac{x^2}{(x + 1)^2}$

II0054

30. $\int \frac{x^2 + 3}{x^6(x^2 + 1)} dx$ equals

- (A) $C - \frac{2}{x} + \frac{2}{3x^3} - \frac{3}{5x^5} - 2 \tan^{-1} x$ (B) $C - \frac{2}{x^2} + \frac{2}{3x^3} - \frac{3}{5x^5} - 2 \tan^{-1} x$
(C) $C - \frac{2}{x} + \frac{2}{3x^2} - \frac{3}{5x^4} - 2 \tan^{-1} x$ (D) $C - \frac{2}{x^2} + \frac{2}{3x^4} + \frac{3}{5x^6} + 2 \tan^{-1} x$

II0055

EXERCISE (O-2)

Multiple Correct Answer Type

1. $\int \frac{5x^4 + 4x^5}{(x^5 + x + 1)^2} dx$ equals

(A) $C - \frac{x+1}{x^5 + x + 1}$

(B) $C + \frac{x}{x^5 + x + 1}$

(C) $C + \frac{x^5}{x^5 + x + 1}$

(D) $C - \frac{x+1}{x^5 - x + 1}$

II0047

2. Which one of the following is **FALSE** ?

(A) $x \cdot \int \frac{dx}{x} = x \ln |x| + C$

(B) $x \cdot \int \frac{dx}{x} = x \ln |x| + Cx$

(C) $\frac{1}{\cos x} \cdot \int \cos x \, dx = \tan x + C$

(D) $\frac{1}{\cos x} \cdot \int \cos x \, dx = x + C$

II0022

3. If $I_n = \int (\sin x)^n dx$ $n \in \mathbb{N}$, then $5I_4 - 6I_6$ is equal to -

(A) $\sin x \cdot (\cos x)^5 + C$

(B) $\cos x \cdot (\sin x)^5 + C$

(C) $\frac{\sin 2x}{8} [\cos^2 2x + 1 - 2 \cos 2x] + C$

(D) $\frac{\sin 2x}{8} [\cos^2 2x + 1 + 2 \cos 2x] + C$

II0023

4. Let $f(x) = \sin^3 x + \sin^3 \left(x + \frac{2\pi}{3} \right) + \sin^3 \left(x + \frac{4\pi}{3} \right)$ then the primitive of $f(x)$ w.r.t. x is

(A) $-\frac{3 \sin 3x}{4} + C$

(B) $\frac{1}{2} \cos^2 \left(\frac{3x}{2} \right) + C$

(C) $\frac{\sin 3x}{4} + C$

(D) $\frac{\cos 3x}{4} + C$

where C is an arbitrary constant.

II0024

5. Suppose $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ and $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. If C is an arbitrary constant of integration then which of the following is/are correct ?

(A) $J = \frac{1}{2}(x - \sin x + \cos x) + C$ (B) $J = K - (\sin x + \cos x) + C$
 (C) $J = x - K + C$ (D) $K = \frac{1}{2}(x - \sin x + \cos x) + C$

II0025

* In each question C is arbitrary constant.

6. The value of $\int 2^{mx} \cdot 3^{nx} dx$ (when $m, n \in \mathbb{N}$) is equal to :

(A) $\frac{2^{mx} + 3^{nx}}{m \ln 2 + n \ln 3} + C$ (B) $\frac{e^{(m \ln 2 + n \ln 3)x}}{m \ln 2 + n \ln 3} + C$
 (C) $\frac{2^{mx} \cdot 3^{nx}}{\ln(2^m \cdot 3^n)} + C$ (D) $\frac{(m \cdot n) \cdot 2^x \cdot 3^x}{m \ln 2 + n \ln 3} + C$

II0104

7. If $f\left(\frac{1-x}{1+x}\right) = x$ and $g(x) = \int f(x) dx$ then

(A) $g(x)$ is continuous in domain (B) $g(x)$ is discontinuous at two points in its domain
 (C) $\lim_{x \rightarrow \infty} g'(x) = -1$ (D) $\int g(x) dx = -\frac{x^2}{2} + (2x+1) \ln\left(\frac{1+x}{e}\right) + C$

II0105

8. If $\int \tan^5 x dx = A \tan^4 x - \frac{1}{2} \tan^2 x + B \ln|\sec x| + C$ then

(A) $A = \frac{1}{4}$ (B) $A = \frac{1}{2}$ (C) $B = 1$ (D) $B = -1$

II0106

9. The value of $\int \{1 + 2 \tan x (\tan x + \sec x)\}^{1/2} dx$ is equal to

(A) $\ln |\sec x (\sec x - \tan x)| + C$ (B) $\ln |\operatorname{cosec} x (\sec x + \tan x)| + C$
 (C) $\ln |\sec x (\sec x + \tan x)| + C$ (D) $-\ln |\cos x (\sec x - \tan x)| + C$

II0107

10. The value of $\int \frac{\ln\left(\frac{x-1}{x+1}\right)}{x^2 - 1} dx$ is equal to

(A) $\frac{1}{2} \ln^2 \frac{x-1}{x+1} + C$ (B) $\frac{1}{4} \ln^2 \frac{x-1}{x+1} + C$ (C) $\frac{1}{2} \ln^2 \frac{x+1}{x-1} + C$ (D) $\frac{1}{4} \ln^2 \frac{x+1}{x-1} + C$

II0108

11. The value of $\int \frac{\ln (\tan x)}{\sin x \cos x} dx$ is equal to

(A) $\frac{1}{2} \ln^2 (\cot x) + C$

(B) $\frac{1}{2} \ln^2 (\sec x) + C$

(C) $\frac{1}{2} \ln^2 (\sin x \sec x) + C$

(D) $\frac{1}{2} \ell n^2 (\cos x \operatorname{cosec} x) + C$

II0109

12. The value of $\frac{\cos^3 x}{\sin^2 x + \sin x} dx$ is equal to :

(A) $\ln |\sin x| + \sin x + C$

(B) $\ln |\sin x| - \sin x + C$

(C) $-\ln |\operatorname{cosec} x| - \sin x + C$

(D) $-\ln |\sin x| + \sin x + C$

II0110

13. If $\int \frac{(x-1) dx}{x^2 \sqrt{2x^2 - 2x + 1}} = \frac{\sqrt{f(x)}}{g(x)} + C$, where $f(x)$ is a quadratic expression and $g(x)$ is a monic linear expression.

(A) $f(x) = 2x^2 - 2x + 1$

(B) $g(x) = x + 1$

(C) $g(x) = x$

(D) $f(x) = 2x^2 - 2x$

II0111

14. If $\int e^{3x} \cos 4x \, dx = e^{3x} (A \sin 4x + B \cos 4x) + C$ then :

(A) $4A = 3B$

(B) $2A = 3B$

(C) $3A = 4B$

(D) $4A + 3B = 1$

II0112

15. $I = \int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx$ equals to

$$(A) -x + \frac{2}{\pi}(2x - 1)\sin^{-1} \sqrt{x} + \frac{2}{\pi}\sqrt{x-x^2} + C$$

$$(B) \ x - \frac{4x}{\pi} \cos^{-1} \sqrt{x} - \frac{2}{\pi} \sin^{-1} \sqrt{x} + \frac{2}{\pi} \sqrt{x} \sqrt{1-x} + C$$

$$(C) -x + \frac{2}{\pi}(2x + 1)\cos^{-1}\sqrt{x} + \frac{2}{\pi}\sqrt{x}\sqrt{1-x} + C$$

(D) $x - \frac{4x}{\pi} \sin^{-1} \sqrt{x} + C$

II0113

16. If $\int \frac{x^2 - x + 1}{(1 + x^2)^{3/2}} e^x dx = e^x f(x) + C$ then

(A) $f(x)$ is an even function

(B) $f(x)$ is a bounded function

(C) Range of $f(x)$ is $(0, 1]$

(D) $f(x)$ has two points of extrema.

II0114

17. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B \ln |9e^{2x} - 4| + C$, then

- (A) $A + 18B = 16$ (B) $18B - A = 19$ (C) $A - 18B = 17$ (D) $A + 18B = 32$

II0115

18. If $\int \frac{dx}{5 + 4\cos x} = P \tan^{-1} \left(m \tan \frac{x}{2} \right) + C$ then :

- (A) $P = 2/3$ (B) $m = 1/3$ (C) $P = 1/3$ (D) $m = 2/3$

II0116

19. The value of $\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$ is equal to :

- (A) $\cot^{-1}(\cot^2 x) + C$ (B) $-\cot^{-1}(\tan^2 x) + C$ (C) $\tan^{-1}(\tan^2 x) + C$ (D) $-\tan^{-1}(\cos 2x) + C$

II0117

20. $\int \frac{x + x^{2/3} + x^{1/6}}{x(1 + x^{1/3})} dx = px^{2/3} + q \tan^{-1}(x^{1/6}) + c$, then :-

- (A) $p < q$ (B) $p = q$ (C) $pq = 9$ (D) $p > q$

II0118

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

A curve is represented parametrically by the equations $x = e^t \cos t$ and $y = e^t \sin t$ where t is a parameter. Then

1. The relation between the parameter ' t ' and the angle α between the tangent to the given curve and the x -axis is given by, ' t ' equals

(A) $\frac{\pi}{2} - \alpha$ (B) $\frac{\pi}{4} + \alpha$ (C) $\alpha - \frac{\pi}{4}$ (D) $\frac{\pi}{4} - \alpha$

II0016

2. The value of $\frac{d^2y}{dx^2}$ at the point where $t = 0$ is

(A) 1 (B) 2 (C) -2 (D) 3

II0016

3. If $F(t) = \int (x + y) dt$ then the value of $F\left(\frac{\pi}{2}\right) - F(0)$ is

(A) 1 (B) -1 (C) $e^{\pi/2}$ (D) 0

II0016

Paragraph for Question 4 to 6

It is known that

$$\sqrt{\tan x} + \sqrt{\cot x} = \begin{cases} \frac{\sqrt{\sin x}}{\sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x}} & \text{if } 0 < x < \frac{\pi}{2} \\ \frac{\sqrt{-\sin x}}{\sqrt{-\cos x}} + \frac{\sqrt{-\cos x}}{\sqrt{-\sin x}} & \text{if } \pi < x < \frac{3\pi}{2} \end{cases}$$

$$\frac{d}{dx}(\sqrt{\tan x} - \sqrt{\cot x}) = \frac{1}{2}(\sqrt{\tan x} + \sqrt{\cot x})(\tan x + \cot x), \quad \forall x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$$

$$\text{and } \frac{d}{dx}(\sqrt{\tan x} + \sqrt{\cot x}) = \frac{1}{2}(\sqrt{\tan x} - \sqrt{\cot x})(\tan x + \cot x), \quad \forall x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right).$$

4. Value of integral $\int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$ is

(A) $\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C$ (B) $\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{2}} \right) + C$

(C) $-\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C$ (D) $-\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{2}} \right) + C$

II0097

5. Value of the integral $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(0, \frac{\pi}{2}\right)$, is

- (A) $\sqrt{2} \sin^{-1} (\cos x - \sin x) + C$ (B) $\sqrt{2} \sin^{-1} (\sin x - \cos x) + C$
 (C) $\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$ (D) $-\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$

II0098

6. Value of the integral $\int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(\pi, \frac{3\pi}{2}\right)$, is

- (A) $\sqrt{2} \sin^{-1} (\cos x - \sin x) + C$ (B) $\sqrt{2} \sin^{-1} (\sin x - \cos x) + C$
 (C) $\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$ (D) $-\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$

II0099

Paragraph for Question 7 and 8

Consider, $\int (\cos^{-1} x + \cos^{-1} \sqrt{1-x^2}) dx = Ax + f(x) \sin^{-1} x + C \sqrt{1-x^2} + D \quad \forall x \in [-1, 0)$

(Where $\pi = 3.14$)

7. The value of $|A + C|$ is II0100

8. If range of $f(x)$ is $(a, b]$ and $\int \sin^{-1}(\cos x) dx = Kx + Fx^2 + G, \forall x \in (a, b]$, then $(K + F)$ is - II0101

Matrix Match Type

9.	List-I	List-II	
(P)	$\int \frac{x^2(x^6 + x^5 - 1)dx}{(2x^6 + 3x^5 + 2)^2}$	(1)	$-\frac{1}{3} \frac{1}{(x^3 - x + 1)} + C$
			II0029
(Q)	$\int \frac{(x^5 + x^4 + x^2)dx}{\sqrt{4x^7 + 5x^6 + 10x^4}}$	(2)	$\frac{1}{2} (1 + x^{-2} + x^{-5})^{-2} + C$
			II0030
(R)	$\int \frac{(2x^{12} + 5x^9)dx}{(x^5 + x^3 + 1)^3}$	(3)	$-\frac{1}{6} (2x^3 + 3x^2 + 2x^{-3})^{-1} + C$
			II0031
(S)	$\int \frac{x^2 - \frac{1}{3}}{(x^3 - x + 1)^2} dx$	(4)	$x \left(\frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$
			II0102

(where C is the constant of integration.)

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$ (B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 0$
 (C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$ (D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

10. Column-I

Column-II

(A) Let $f(x) = \int x^{\sin x} (1 + x \cos x \cdot \ln x + \sin x) dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$

(P) rational

then the value of $f(\pi)$ is

II0026

(B) Let $g(x) = \int \frac{1 + 2 \cos x}{(\cos x + 2)^2} dx$ and $g(0) = 0$

(Q) irrational

then the value of $g\left(\frac{\pi}{2}\right)$ is

II0027

(C) Let $k(x) = \int \frac{(x^2 + 1) dx}{\sqrt[3]{x^3 + 3x + 6}}$ and $k(-1) = \frac{1}{\sqrt[3]{2}}$ then the value

(R) integral

of $k(-2)$ is

II0103

(D) If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln(f(x)) + g(x) + C$

(S) prime

(where C is the constant of integration and $f(x)$ is positive),

then $f(0) + g(0)$ is

II0028

EXERCISE (O-4)

Numerical Grid Type

1. Let $\int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x) + g(x))\sqrt{f(x)g(x) - g^2(x)}} dx = \sqrt{m} \tan^{-1} \left(\sqrt{\frac{f(x) - g(x)}{ng(x)}} \right) + C$,
where $m, n \in \mathbb{N}$ and 'C' is constant of integration ($g(x) > 0$). Find the value of $(m^2 + n^2)$. II0034
2. A function g defined for all positive real numbers, satisfies $g'(x^2) = x^3$ for all $x > 0$ and $g(1) = 1$. Compute $g(4)$. II0041
3. If the value $\int \frac{1 - (\cot x)^{2008}}{\tan x + (\cot x)^{2009}} dx = \frac{1}{k} \ln |\sin^k x + \cos^k x| + C$, then find k . II0050
4. Suppose $\int \frac{1 - 7 \cos^2 x}{\sin^7 x \cos^2 x} dx = \frac{g(x)}{\sin^7 x} + C$, where C is arbitrary constant of integration. Then find the value of $g'(0) + g''\left(\frac{\pi}{4}\right)$ II0051
5. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x)dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$ II0069
6. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln |\sin x - 2 \cos x| + k$ then a is equal to : II0071
7. If $\int \frac{\cos^5 x}{\sin x} dx = \ell \log |\sin x| + m \sin^4 x + n \sin^2 x + C$, (where C is constant of integration), then $\frac{\ell - n}{m}$ is II0119
8. If $\int \frac{x^4 - 2x^2 - 3}{x^4 \sqrt{x(1+x^{-2})^2 + 1}} dx = 2(a + bx + cx^{-1} + dx^{-3})^{\frac{1}{2}} + K$, where K is the integration constant, then the value of $a + b + c + d$ is II0120
9. Let F is anti-derivative of f defined by $f(x) = 5(x^4 - 1)$ where $F(1) = 20$, then value of $F(2)$ is II0121
10. $\int \frac{dx}{16 + x^2} = f(x)$ where $f(4) = \frac{\pi}{16}$ and $(\pi = 3.14)$, then value of $f(2) + f\left(\frac{4}{3}\right)$ is - II0122

EXERCISE (JM)

1. If $\int f(x)dx = \Psi(x)$, then $\int x^5 f(x^3)dx$ is equal to : [JEE-MAIN-2013]

- (1) $\frac{1}{3} \left[x^3 \Psi(x^3) - \int x^2 \Psi(x^3) dx \right] + C$ (2) $\frac{1}{3} x^3 \Psi(x^3) - 3 \int x^3 \Psi(x^3) dx + C$
 (3) $\frac{1}{3} x^3 \Psi(x^3) - \int x^2 \Psi(x^3) dx + C$ (4) $\frac{1}{3} \left[x^3 \Psi(x^3) - \int x^3 \Psi(x^3) dx \right] + C$

II0072

2. The integral $\int \left(1 + x - \frac{1}{x} \right) e^{\frac{x+1}{x}} dx$ is equal to : [JEE-MAIN-2014]

- (1) $(x-1)e^{\frac{x+1}{x}} + c$ (2) $xe^{\frac{x+1}{x}} + c$ (3) $(x+1)e^{\frac{x+1}{x}} + c$ (4) $-xe^{\frac{x+1}{x}} + c$

II0073

3. The integral $\int \frac{dx}{x^2(x^4+1)^{\frac{3}{4}}}$ equals : [JEE-MAIN-2015]

- (1) $-(x^4+1)^{\frac{1}{4}} + c$ (2) $-\left(\frac{x^4+1}{x^4}\right)^{\frac{1}{4}} + c$ (3) $\left(\frac{x^4+1}{x^4}\right)^{\frac{1}{4}} + c$ (4) $(x^4+1)^{\frac{1}{4}} + c$

II0074

4. The integral $\int \frac{2x^{12}+5x^9}{(x^5+x^3+1)^3} dx$ is equal to :- [JEE-MAIN 2016]

- (1) $\frac{-x^{10}}{2(x^5+x^3+1)^2} + C$ (2) $\frac{-x^5}{(x^5+x^3+1)^2} + C$ (3) $\frac{x^{10}}{2(x^5+x^3+1)^2} + C$ (4) $\frac{x^5}{2(x^5+x^3+1)^2} + C$

where C is an arbitrary constant.

II0075

5. Let $I_n = \int \tan^n x dx, (n > 1)$. $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to :- [JEE-MAIN 2017]

- (1) $\left(-\frac{1}{5}, 0\right)$ (2) $\left(-\frac{1}{5}, 1\right)$ (3) $\left(\frac{1}{5}, 0\right)$ (4) $\left(\frac{1}{5}, -1\right)$

II0076

6. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is equal to [JEE-MAIN 2018]

- (1) $\frac{-1}{3(1+\tan^3 x)} + C$ (2) $\frac{1}{1+\cot^3 x} + C$ (3) $\frac{-1}{1+\cot^3 x} + C$ (4) $\frac{1}{3(1+\tan^3 x)} + C$

(where C is a constant of integration)

II0077

7. For $x^2 \neq n\pi + 1$, $n \in \mathbb{N}$ (the set of natural numbers), the integral $\int x \sqrt{\frac{2\sin(x^2 - 1) - \sin 2(x^2 - 1)}{2\sin(x^2 - 1) + \sin 2(x^2 - 1)}} dx$ is

equal to :

(where c is a constant of integration)

(1) $\log_e \left| \sec \left(\frac{x^2 - 1}{2} \right) \right| + c$

(2) $\log_e \left| \frac{1}{2} \sec^2(x^2 - 1) \right| + c$

(3) $\frac{1}{2} \log_e \left| \sec^2 \left(\frac{x^2 - 1}{2} \right) \right| + c$

(4) $\frac{1}{2} \log_e |\sec(x^2 - 1)| + c$ [JEE-MAIN 2019]

II0078

8. If $\int \frac{\sqrt{1-x^2}}{x^4} dx = A(x)(\sqrt{1-x^2})^m + C$, for a suitable chosen integer m and a function $A(x)$, where C is a constant of integration then $(A(x))^m$ equals :

[JEE-MAIN 2019]

(1) $\frac{-1}{3x^3}$

(2) $\frac{-1}{27x^9}$

(3) $\frac{1}{9x^4}$

(4) $\frac{1}{27x^6}$

II0079

9. The integral $\int \cos(\log_e x) dx$ is equal to : (where C is a constant of integration) [JEE-MAIN 2019]

(1) $\frac{x}{2} [\sin(\log_e x) - \cos(\log_e x)] + C$

(2) $\frac{x}{2} [\cos(\log_e x) + \sin(\log_e x)] + C$

(3) $x [\cos(\log_e x) + \sin(\log_e x)] + C$

(4) $x [\cos(\log_e x) - \sin(\log_e x)] + C$

II0080

10. If $\int \frac{\cos x dx}{\sin^3 x (1 + \sin^6 x)^{2/3}} = f(x)(1 + \sin^6 x)^{1/\lambda} + c$

where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to

[JEE-MAIN 2020]

(1) -2

(2) $-\frac{9}{8}$

(3) 2

(4) $\frac{9}{8}$

II0081

11. If $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \log_e |f(\theta)| + C$ where C is a constant of integration, then the ordered pair $(\lambda, f(\theta))$ is equal to : [JEE-MAIN 2020]

(1) $(-1, 1 + \tan \theta)$ (2) $(-1, 1 - \tan \theta)$ (3) $(1, 1 - \tan \theta)$ (4) $(1, 1 + \tan \theta)$

II0082

12. The integral $\int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}}$ is equal to :

(where C is a constant of integration)

[JEE-MAIN 2020]

(1) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$ (2) $-\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$ (3) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}} + C$ (4) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{\frac{13}{7}} + C$

II0083

13. If $\int \frac{\cos x - \sin x}{\sqrt{8 - \sin 2x}} dx = a \sin^{-1} \left(\frac{\sin x + \cos x}{b} \right) + c$, where c is a constant of integration, then the ordered pair (a, b) is equal to : [JEE-MAIN 2021]

(1) $(-1, 3)$ (2) $(3, 1)$ (3) $(1, 3)$ (4) $(1, -3)$ II0123

14. $\int \frac{(x^2 + 1)e^x}{(x+1)^2} dx = f(x)e^x + C$, Where C is a constant, then $\frac{d^3 f}{dx^3}$ at $x = 1$ is equal to :

(A) $-\frac{3}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$ II0124

[JEE-MAIN 2022]

15. For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x} dx$ if $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then [JEE-MAIN 2022]

(A) $3^{1010} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ (B) $3^{1010} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

(C) $3^{1011} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ (D) $3^{1011} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$ II0125

5. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K) [JEE 2012, 3M, -1M]

(A) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

(B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

(C) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

(D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

II0088

ANSWER KEY

Do yourself - 1

1. $\frac{2^x \cdot e^x}{1 + \ln 2} + C$
2. $\frac{x^{a+1}}{a+1} + \frac{a^x}{\ln a} + C$
3. $\frac{x^3}{3} + C$
4. $\frac{x^5}{5} - \frac{x^3}{3} + x - 2 \tan^{-1} x + C$
5. $\frac{1}{2}(\tan x + x) + C$
6. $-(\cot x + \tan x) + C$
7. $\tan x - x + C$
8. $x + C$
9. $2(\sin x + x \cos \alpha) + C$
10. $\sec x - \operatorname{cosec} x + C$
11. $\frac{x}{\sqrt{2}} + C$
12. $\tan x - x + C$
13. $-\frac{1}{x} + \tan^{-1} x + C$
14. $\ln|x| + 2 \tan^{-1} x + C$
15. $2x + 3 \ln|x-2| + C$
16. $-\frac{2}{x} + \tan^{-1} x + C$
17. $\frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2} \ln(x^2 + 1) + 3 \tan^{-1} x + C$
18. $\frac{1}{2} \left[\frac{x^3}{3} + \tan^{-1} x \right] + C$
19. $\tan x - \tan^{-1} x + C$
20. $\tan x - \cot x - 3x + C$
21. $-2 \cos x + C$
22. $\tan x - \sec x + C$
23. $(\sin x - \cos x) + (\sin k + \cos k)x + C$
24. $C - \cot \frac{x}{2}$

Do yourself - 2

1. $C - \frac{e^{1-3x}}{3}$
2. $\frac{(x+1)^{16}}{16} + C$
3. $e^x + e^{-x} + C$
4. $C - \frac{1}{8(2x-3)^4}$
5. $\frac{(a+bx)^{1-c}}{b(1-c)} + C$
6. $C - \frac{5}{33}(8-3x)^{11/5}$
7. $C - \frac{\sqrt{(8-2x)^3}}{3}$
8. $\frac{3m}{b} \sqrt[3]{a+bx} + C$
9. $\frac{2}{3}x + \frac{5}{9} \ln(3x+2) + C$
10. $\frac{1}{3}e^{3x} + \frac{3}{2}e^{2x} + 3e^x + x + C$
11. $\frac{x}{2} + \frac{\sin 2x}{4} + C$
12. $\frac{x}{2} - \frac{\sin 2x}{4} + C$
13. $2 \tan \frac{x}{2} - x + C$
14. $C - \frac{1}{4} \left(\frac{\cos 4x}{2} + \cos 2x \right)$
15. $x \cos \alpha - \frac{1}{2} \sin 2x + C$
16. $\frac{1}{2} \tan \left(2x - \frac{\pi}{4} \right) + C$ or $\frac{1}{2}(\tan 4x - \sec 4x) + C$
17. $\frac{1}{8} \left(2x + \sin 2x + \frac{1}{2} \sin 4x + \frac{1}{3} \sin 6x \right) + C$
18. $c - 2 \cot 2\phi$
19. $\frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$
20. $\frac{1}{2} \sin 2x + C$

Do yourself - 4

1. $C - \cos(e^x)$
2. $\frac{2}{3}(x+2)^{3/2} - 4(x+2)^{1/2} + C$
3. $2\ln(e^{x/2} + e^{-x/2}) + C$
4. $x^x + C$
5. $x^2\sqrt{1+x^2} - \frac{2}{3}\sqrt{(1+x^2)^3} + C$
6. $\frac{1}{2}\left[x - \frac{\ln(2x+1)}{2}\right] + C$
7. $C - \frac{1}{4}x^4 - \frac{1}{3}x^3 - \frac{1}{2}x^2 - x - \ln|1-x|$
8. $\frac{x^3}{3} - \frac{x^2}{2} + x - \ln|x+1| + C$
9. $2\left[\sqrt{x+1} - \ln(1+\sqrt{x+1})\right] + C$
10. $\frac{3}{2}(x+1)^{2/3} - 3(x+1)^{1/3} + 3\ln|1+\sqrt[3]{x+1}| + C$
11. $\frac{(\sqrt{1+x^2}+x)^{15}}{15} + C$
12. $\frac{x^2}{2} - x + C$
13. $6\left[\frac{t^4}{4} - \frac{t^2}{2} + t + \frac{1}{2}\ln(1+t^2) - \tan^{-1}t\right] + C$ where $t = x^{1/6}$
14. $\ln|\ln(\sin x)| + C$
15. $\frac{\sqrt{4+x^2}(x^2-2)}{24x^3} + C$
16. $\frac{\sqrt{x^2-3}(2x^2+3)}{27x^3} + C$
17. $\frac{2}{3}[x^3 - \sqrt{(x^2-1)^3}] - x + C$

Do yourself - 5

1. (a) $\frac{1}{4}\sin^4 x + C$ (b) $\sec x + C$
2. (a) $\frac{1}{\cos x - 1} + C$ (b) $\frac{(\arctan x)^3}{3} + C$
3. (a) $\sin x - \frac{1}{3}\sin^3 x + C$ (b) $3\sqrt[3]{\sin x} + C$ (c) $C - \frac{2}{5}\cos^5 x$
- (d) $\frac{1}{2}(\tan x)^2 + C$ or $\frac{\sec^2 x}{2} + C$
4. $2\sqrt{1+\tan x} + C$
5. $C - \ln(1+\cos^2 x)$
6. $e^{\sin x} + C$
7. $\frac{1}{3}\tan^3 x + C$
8. (a) $\frac{1}{2}\tan^2 x + \ln|\cos x| + C$ (b) $\frac{1}{3}\tan^3 x - \tan x + x + C$
9. $\ln(1+\sin x) + C$
10. $\tan x + \frac{1}{3}\tan^3 x + C$
11. (a) $\frac{\cos^2 x}{2} - \ln|\cos x| + C$ (b) $\frac{1}{\sin x} - \frac{1}{3\sin^3 x} + C$

12. $2\sqrt{\cos\alpha}\left(\frac{\cos^2\alpha}{5}-1\right)+C$ 13. $C-\frac{1}{2(\arcsin x)^2}$ 14. $\tan\left(\frac{x}{2}-\frac{\pi}{4}\right)+C$
15. $2\tan\left(\frac{x}{2}+\frac{\pi}{4}\right)-x+C$ 16. $\ell n\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right|+C$
17. (a) $\ell n|x+\cos x|+C$ (b) $\frac{1}{2}\ell n|x^2+\sin 2x+2x|+C$
19. $\frac{1}{2}\ell n\left|\tan\left(\frac{\phi}{2}+\frac{\pi}{6}\right)\right|+c$ 20. $2\ell n\left(\sec\frac{x}{2}\right)-3\ell n\left(\sec\frac{x}{3}\right)-6\ell n\left(\sec\frac{x}{6}\right)+C$
21. $\ell n\left(\frac{\sec(\ell n x)}{\sec(\ell n x/2)x^{\tan(\ell n 2)}}\right)+C$ 22. $\ell n\left(\frac{\sin x}{x}\right)+C$ 23. $\frac{1}{2}x-\frac{1}{2}\ell n|\sin x+\cos x|+C$

Do yourself - 6

1. $\frac{1}{3\sqrt{2}}\arctan\frac{\sqrt{2}}{3}x+C$ 2. $\arcsin\frac{x}{2}+C$ 3. $\frac{1}{3}\arctan 3x+C$
4. $\frac{1}{5}\arcsin 5x+C$ 5. $\frac{1}{4}\sin^{-1}\frac{4}{3}x+C$ 6. $\frac{1}{10}\tan^{-1}\frac{2x}{5}+C$
7. $\frac{1}{36}\tan^{-1}\left(\frac{4x^3}{3}\right)+C$ 8. $\frac{1}{2}\ell n\left[\frac{\sqrt{x^2+4}-2}{x}\right]+C$ 9. $\frac{1}{3}\arcsin\frac{3x}{2}+C$
10. $\arcsin(x-2)+C$ 11. $\frac{1}{a}\arctan\frac{\sin\alpha}{a}+C$ 12. $\frac{1}{2}\arctan x^2+C$
13. $\frac{1}{2}\arcsin\frac{x^2}{a}+C$ 14. $\frac{1}{6}\arctan\frac{x^3}{2}+C$ 15. $\frac{1}{4}\arcsin x^4+C$
16. $\frac{1}{2}\arctan\frac{e^x}{2}+C$ 17. $\frac{\arcsin 2^x}{\ell n 2}+C$ 18. $\arcsin x-\sqrt{1-x^2}+C$
19. $\frac{3}{2}\ell n(x^2+9)-\frac{1}{3}\arctan\frac{x}{3}+C$ 20. $\arcsin x+\sqrt{1-x^2}+C$ 21. $\sqrt{(x-2)(3-x)}-\sin^{-1}\sqrt{3-x}+C$
22. $\frac{1}{2}\arctan x^2-\frac{1}{4}\ell n(x^4+1)+C$ 23. $C-2\sqrt{1-x^2}-\frac{2}{3}\sqrt{(\arcsin x)^3}$ 24. $C-\frac{1}{9}[\sqrt{1-9x^2}+(\arccos 3x)^3]$
25. $\arcsin x+\frac{1}{\sqrt{1-x^2}}+C$ 26. $\frac{1}{2}\arctan\frac{x-1}{2}+C$ 27. $\frac{1}{2}\arcsin(2x+3)+C$
28. $2\sqrt{1+x^2}+3\ell n(x+\sqrt{1+x^2})+c$ 29. $\arcsin\frac{\ell nx}{\sqrt{3}}+c$
30. $\frac{1}{2\sqrt{2}}\ell n\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right|+\frac{1}{2\sqrt{3}}\left|\frac{x-\sqrt{3}}{x+\sqrt{3}}\right|+c$ 31. $\int\frac{dx}{4x^2-9}$ 32. $\int\frac{dx}{2-3x^2}$

Do yourself - 7

1. $\frac{1}{\sqrt{2}} \arctan \frac{x+1}{\sqrt{2}} + C$
2. $\frac{2}{3} \arctan \frac{1-2x}{3} + C$
3. $\frac{1}{4} \arctan \frac{2x+1}{2} + C$
4. $\frac{1}{3} \arcsin \frac{3x-1}{3} + C$
5. $\frac{1}{3} \arcsin \frac{3x+1}{\sqrt{3}} + C$
6. $\ln \sqrt{\frac{x^2-2}{x^2-1}} + C$
7. $\frac{1}{x} + \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + c$
8. $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$
9. $2 \arcsin \sqrt{x} + C$
10. $\frac{1}{3} \ln \left| \frac{x-5}{x-2} \right| + C$
11. $\frac{1}{\sqrt{x^2+1}} + \ln(x + \sqrt{x^2+1}) + c$
12. $\frac{1}{2} \ln(x^2 + 2x + 2) + \arctan(x+1) + c$
13. $c - \sqrt{3-2x-x^2} - 4 \arcsin \frac{x+1}{2}$
14. $3\sqrt{x^2+2x+2} - 4 \ln(x+1 + \sqrt{x^2+2x+2}) + c$
15. $\frac{1}{7} \ln \left| \frac{x-2}{x+5} \right| + C$
16. $5\sqrt{x^2+4x+1} - 6 \ln \left[(x+2) + \sqrt{x^2+4x+1} \right] + C$

Do yourself - 8

1. $\frac{x^2}{2} \ln x - \frac{x^2}{4} + C$
2. $C - e^{-x}(x+1)$
3. $\frac{x^2}{4} - \frac{x}{4} \sin 2x - \frac{1}{8} \cos 2x + C$
4. $(x^2-1)e^{x^2} + C$
5. $xe^x - e^x + C$
6. $\frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + C$
7. $x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) - \frac{(\tan^{-1} x)^2}{2} + C$
8. $\frac{(x^2+1) \arctan x}{\sqrt{x}} - 2\sqrt{x} + C$
9. $\frac{x^{n+1}}{n+1} \left(\ln x - \frac{1}{n+1} \right) + C$
10. $2(\sqrt{x} - \sqrt{1-x} \arcsin \sqrt{x}) + C$
11. $\sqrt{1+x^2} \arctan x - \ln(x + \sqrt{1+x^2}) + C$
12. $C - \frac{1}{x} (\ln^3 x + 3 \ln^2 x + 6 \ln x + 6)$
13. $C - \frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \arcsin x$
14. $x \tan^{-1} \sqrt{x} - \sqrt{x} + \arctan \sqrt{x} + C$

15. $2\sqrt{e^x - 1} - 2\arctan \sqrt{e^x - 1} + C$

16. $x \ln(x + \sqrt{1 + x^2}) - \sqrt{1 + x^2} + C$

17. $\frac{x}{2}(\sin \ln x - \cos \ln x) + C$

18. $\frac{e^{ax}}{a^2 + n^2} (n \sin nx + a \cos nx) + C$

19. $\frac{x^2+1}{2}(\arctan x)^2 - x \arctan x + \frac{1}{2} \ln(1+x^2) + C$

Do yourself - 9

1. $e^x \tan^{-1} x + C$

2. $\frac{1}{2}e^{x^2} \sin(x^2) + C$

3. $x \tan(e^x) + C$

4. $x \tan \frac{x}{2} + C$

5. $x \ln x + C$

6. $e^{\sin x} (x - \sec x) + c$

7. $C - e^{\cos x}(x + \operatorname{cosec} x)$

8. (D)

9. (D)

10. D

11. B

12. C

13. C

14. C

15. D

Do yourself - 10

1. $C - \cot x - \frac{2}{3}\cot^3 x - \frac{1}{5}\cot^5 x$

2. $C - \cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x$

3. $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$

4. $\frac{1}{3}\tan^3 x + C$

5. $\frac{1}{3}\sin^3 x - \frac{2}{5}\sin^5 x + \frac{1}{7}\sin^7 x + C$

6. $\ln |\tan x| - \frac{1}{2 \sin^2 x} + c$

7. $\frac{1}{2}(\tan^2 x - \cot^2 x) + 2 \ln |\tan x| + c$

8. $-\frac{3(1+4\tan^2 x)}{8(\tan x)^{8/3}} + C$

9. $\frac{2}{3}\tan^{3/2} x + C$

10. $\frac{3}{55} \sqrt[3]{\tan^5 x} (5 \tan^2 x + 11) + c$

11. $4\sqrt[4]{\tan x} + c$

12. $\frac{\sqrt{2}}{5}(\tan^2 x + 5)\sqrt{\tan x} + C$

Do yourself - 11

1. $\frac{1}{\sqrt{5}} \tan^{-1}(\sqrt{5} \tan x) + C$

2. $\frac{1}{3} \arctan(3 \tan x) + C$

3. $\frac{2}{\sqrt{15}} \tan^{-1} \left(\frac{8 \tan x + 1}{\sqrt{15}} \right) + C$

4. $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{3 \tan x / 2 + 1}{2\sqrt{2}} \right) + C$

5. $\frac{1}{2 - \tan \frac{x}{2}} + c$

6. $\frac{1}{2\sqrt{6}} \ell_n \left| \frac{\sqrt{6} + \tan x / 2 - 2}{\sqrt{6} - \tan x / 2 + 2} \right| + C$

7. $\frac{12}{13}x - \frac{5}{13} \ln |3 \cos x + 2 \sin x| + C$

8. $\ln(2 + \sin 2x) + C$

9. $\arctan(\tan^2 x) + c$ or $-\tan^{-1}(\cos 2x) + c$

10. $\frac{\cos 2x - 15}{15(4 + \sin 2x)} + \frac{4}{15\sqrt{15}} \arcsin \frac{4 \sin 2x + 1}{4 + \sin 2x} + C$

or

$$\frac{z}{8a^2(z^2 + a^2)} + \frac{1}{8a^3} \tan^{-1} \frac{z}{a} + c, \text{ where } z = \tan x + \frac{1}{4} \text{ \& } a^2 = \frac{15}{16}$$

Do yourself - 12

1. $C - \frac{1}{8} \ln \frac{2 + \cos 2x}{2 - \cos 2x}$

2. $-\frac{1}{2} \ln |x+1| + \frac{7}{2} \ln |x+3| + C$

3. $\ln |x+2| + \frac{3}{x+2} + C$

4. (i) $\ln \left| \frac{x+1}{x+2} \right| + C$

(ii) $\frac{1}{10} \ln |x+3| - \frac{1}{20} \ln |x^2 + 1| + \frac{3}{10} \tan^{-1} x + C$

(iii) $4 \ln |x+1| + \frac{1}{(x+1)} - 4 \ln |x+2| + C$

(iv) $\frac{1}{2} \ln |x+1| - \ln |x+2| + \frac{1}{2} \ln |x+3| + C$

5. $\frac{2}{7} \ln |(x-2)| + \frac{5}{7} \ln |(x+5)| + C$

6. $\frac{3}{11} \ln |3x+1| + \frac{2}{33} + \ln |2x-3| - \frac{1}{3} \ln |x| + c$

7. $\frac{1}{4} \ln \left| \frac{1+x}{1-x} \right| - \frac{1}{2} \arctan x + c$

8. $\ln \left| \frac{(x-1)^4 (x-4)^5}{(x+3)^7} \right| + c$

9. $\frac{1}{2} \tan x + \frac{1}{2\sqrt{2}} \arctan(\sqrt{2} \tan x) + C$

10. $\frac{1}{3} \ln \frac{|x-1|}{\sqrt{x^2+x+1}} + \frac{1}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} + c$

11. $\frac{1}{4} x + \ln |x| - \frac{7}{16} \ln |2x-1| - \frac{9}{16} \ln |2x+1| + c$

12. $2 \ln \left| \frac{x+4}{x+2} \right| - \frac{5x+12}{x^2+6x+8} + c$

13. $-\frac{1}{(x+1)} + \frac{1}{2(x+1)^2} + C$

14. $\frac{2x}{x^3-1} = \frac{2}{3} \frac{1}{x-1} + \frac{2}{3} \frac{1-x}{x^2+x+1}$

15. $\frac{1}{15} \left[\frac{1}{2} \ln \frac{(z-1)^2}{z^2+z+1} - \sqrt{3} \arctan \frac{2z+1}{\sqrt{3}} \right] + c, \text{ where } z = x^5$

16. $C - \frac{11}{2(x-2)^2} - \frac{4}{x-2}$

17. $\frac{x^3}{3} + \frac{x^2}{2} + 4x + \ln \left| \frac{x^2(x-2)^5}{(x+2)^3} \right| + c$

18. $4\ln|x| - 3\ln|x-1| - \frac{9}{x-1} + C$

19. $\frac{x}{a^2\sqrt{x^2+a^2}} + c$

20. $\ln \left| \frac{xe^x}{1+xe^x} \right| + C$

21. $x + \frac{6\sqrt{x^5}}{5} + \frac{3\sqrt[3]{x^2}}{2} + 2\sqrt{x} + 3\sqrt[3]{x} + 6\sqrt[6]{x} + 6\ln|\sqrt[6]{x}-1| + C$

22. $C - \frac{x(x^2-3)}{2\sqrt{1-x^2}} - \frac{3}{2}\arcsin x$

Do yourself - 13

1. $\tan^{-1}\left(\frac{x^2-1}{x}\right) + C$

2. $\frac{1}{2\sqrt{2}} \left[\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right) - \frac{1}{2} \ln \frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1} \right]$

3. $\arctan(e^x - e^{-x}) + C$

4. $\log(\tan^{-1}(\cos x + \sec x)) + c$

5. $\ln \frac{x^2+1+\sqrt{x^4+3x^2+1}}{x} + C$

Do yourself - 14

1. $2\sqrt{x+1} + \frac{3}{2} \ln \left| \frac{\sqrt{x+1}-2}{\sqrt{x+1}+2} \right| + C$

2. $-\frac{1}{x} \sqrt{1+x^2} + C$

3. $\ln \left| \frac{\sqrt{x+2}-1}{\sqrt{x+2}+1} \right| + C$

4. $\frac{1}{4\sqrt{3}} \ln \left| \frac{t-\sqrt{3}}{t+\sqrt{3}} \right| - \frac{1}{2} \tan^{-1}(t) + C$, where $t = \sqrt{x+1}$

5. $\ln \frac{|x|}{1+\sqrt{x^2+1}} + c$

6. $\frac{\sqrt{x^2-9}}{9x} + c$

7. D

8. A

9. B

10. $\frac{\ln(|\sqrt{x+1}-\sqrt{2}|) - \ln(\sqrt{x+1}+\sqrt{2})}{2^{3/2}} + \frac{1}{\sqrt{x+1}} + c$

11. $\frac{1}{2\sqrt{3}} \ln \left(\frac{\sqrt{x+2}+\sqrt{3}}{\sqrt{x+2}-\sqrt{3}} \right) + \frac{1}{2} \ln \left(\frac{|\sqrt{x+2}-1|}{\sqrt{x+2}+1} \right) + c$

12. $\frac{1}{3} \sin^{-1} \left(\frac{3}{2x+3} \right) + c$

Do yourself - 15

- $\frac{1}{4} \sin^{-1} 2x + \frac{1}{2} x \sqrt{1-4x^2} + C$
- $\frac{(x+2)}{2} \sqrt{x^2+4x+6} + \log|x+2+\sqrt{x^2+4x+6}| + C$
- $\frac{5}{2} \sin^{-1} \left(\frac{x+2}{\sqrt{5}} \right) + \frac{x+2}{2} \sqrt{1-4x-x^2} + C$
- $\frac{(2x-3)}{4} \sqrt{1+3x-x^2} + \frac{13}{8} \sin^{-1} \left(\frac{2x-3}{\sqrt{13}} \right) + C$
- $\frac{1}{3} (x^2+x)^{3/2} - \frac{(2x+1)\sqrt{x^2+x}}{8} + \frac{1}{16} \log \left| x + \frac{1}{2} + \sqrt{x^2+x} \right| + C$
- $\frac{1}{6} (2x^2+3)^{3/2} + \frac{x}{2} \sqrt{2x^2+3} + \frac{3\sqrt{2}}{4} \log \left| x + \sqrt{x^2 + \frac{3}{2}} \right| + C$
- $-\frac{1}{3} (3-4x-x^2)^{3/2} + \frac{7}{2} \sin^{-1} \left(\frac{x+2}{\sqrt{7}} \right) + \frac{(x+2)\sqrt{3-4x-x^2}}{2} + C$

Do yourself - 16

- $-\frac{1}{2} \ln \left(\frac{x^2+1}{x^2} \right) + C$
- $-\left(1 + \frac{1}{x^3} \right)^{1/3} + C$
- $-\frac{1}{2} \left(1 + \frac{1}{x^3} \right)^{2/3} + C$
- (D)
- $\frac{1}{\sqrt{\frac{1}{x^2} - x^2}} + c$
- $\frac{2\sqrt{x^2-2x+2}}{x} + c$
- $\frac{-\ln \left(\frac{1}{x^{100}} + 1 \right)}{100} + c$
- $\frac{-(1+x^5)^{1/5}}{x} + c$
- $2 \frac{1}{\sqrt{1 + \frac{\sin \alpha}{x} + \frac{1}{x^2}}} + c$
- $\frac{-(x^{10}+10)^{9/10}}{90x^9} + c$

Do yourself - 17

- $\left(\frac{x}{e} \right)^x - \left(\frac{e}{x} \right)^x + C$
- $\ln \left(\frac{1 + \sqrt{1+3\cos^2 2\theta}}{\cos 2\theta} \right) + C$
- $\sin^{-1} \left(\frac{1}{2} \sec^2 \frac{x}{2} \right) + C$
- $\frac{4}{3} \sqrt[4]{\frac{x-1}{x+2}} + C$
- $\frac{1}{2} \cot \frac{x}{2} - \frac{1}{6} \cot^3 \frac{x}{2} + C$

6. $C - \frac{1}{3} \tan x (2 + \tan^2 x) \sqrt{4 - \cot^2 x}$
7. $\frac{4}{\sqrt{\cos \frac{x}{2}}} + 2 \arctan \sqrt{\cos \frac{x}{2}} - \ln \frac{1 + \sqrt{\cos \frac{x}{2}}}{1 - \sqrt{\cos \frac{x}{2}}} + C$
8. $C - \frac{x}{2(1+x^2)} + \frac{1}{2} \arctan x$
9. $\frac{15x^5 + 40x^3 + 33x}{48(1+x^2)^3} + \frac{15}{48} \arctan x + C$
10. $C - \frac{\sqrt{(1+x^2)^3}}{3x^3}$
11. $\frac{1}{\sqrt{2}} \arccos \frac{x\sqrt{2}}{x^2+1} + C$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	B	B	B	A	B	D	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	A	C	A	C	B	C	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	D	A	B	D	A	C	A

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	A,C,D	B,C	B,D	B,C	B,C	A,C	A,C	C,D	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C,D	B,C	A,C	C,D	A,B	A,B,C	A,B	A,B	A,B,C,D	A,C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	B	C	A	B	A	0.43	1.07	B	
Q.10	A	B	C	D						
	Q	P	P,R,S	P,R						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	8	13.40	2010	5	3	2	8	5	46.00	0.19 or 0.20

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	3	3	1	1 or 3	2	2	1
Que.	11	12	13	14	15					
Ans.	1	1	3	B	A					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5					
Ans.	D	A	D	C	C					

EXERCISE (O-1)

Straight Objective Type

1. The value of $\int_{-1}^1 \frac{dx}{\sqrt{|x|}}$ is -

(A) $\frac{1}{2}$

(B) 2

(C) 4

(D) undefined

DI0002

2. The value of $\int_{\pi}^{2\pi} [2 \cos x] dx$ where $[.]$ represents the greatest integer function, is -

(A) $-\frac{5\pi}{6}$

$$(B) -\frac{\pi}{2}$$

(C) $-\pi$

(D) none

DI0031

3. The value of the definite integral $\int_{19}^{37} (\{x\}^2 + 3(\sin 2\pi x)) dx$ where $\{x\}$ denotes the fractional part function.

(A) 0

(B) 6

(C) 9

(D) can't be determined

DI0013

4. $\int_{-\frac{1}{2}}^{\frac{3}{2}} \left\{ \frac{1}{2} (|x-3| + |1-x|-4) \right\} dx$ equals -

(A) $-\frac{3}{2}$

(B) $\frac{9}{8}$

(C) $\frac{1}{4}$

(D) $\frac{3}{2}$

Where $\{.\}$ denotes the fraction part function.

DI0001

5. $\int_{-1}^1 \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} dx = a \ln 2 + b$ then -

(A) $a = 2; b = 1$

(B) $a = 2; b = 0$

(C) $a = 3$; $b = -2$

(D) $a = 4$; $b = -1$

DI0015

6. The value of the definite integral $\int_2^4 (x(3-x)(4+x)(6-x)(10-x) + \sin x) dx$ equals -

(A) $\cos 2 + \cos 4$

(B) $\cos 2 - \cos 4$

(C) $\sin 2 + \sin 4$

(D) $\sin 2 - \sin 4$

DI0003

7. Value of the definite integral $\int_{-1/2}^{1/2} (\sin^{-1}(3x - 4x^3) - \cos^{-1}(4x^3 - 3x)) dx$

- (A) 0 (B) $-\frac{\pi}{2}$ (C) $\frac{7\pi}{2}$ (D) $\frac{\pi}{2}$

DI0005

8. The value of the definite integral $\int_0^{\pi/2} \sin x \sin 2x \sin 3x dx$ is equal to :

- (A) $\frac{1}{3}$ (B) $-\frac{2}{3}$ (C) $-\frac{1}{3}$ (D) $\frac{1}{6}$

DI0006

9. If $I = \int_0^{\pi/2} \ln(\sin x) dx$ then $\int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x) dx$

- (A) $\frac{I}{2}$ (B) $\frac{I}{4}$ (C) $\frac{I}{\sqrt{2}}$ (D) I

DI0011

10. $\int_0^{\pi/2} \frac{\cos \theta - \sin \theta}{(1 + \cos \theta)(1 + \sin \theta)} d\theta$ equals -

- (A) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\cos^{-1}(0)$ (C) $\cos^{-1}(1)$ (D) $\cos^{-1}(-1)$

DI0019

11. Let $f(x)$ be a continuous function on $[0, 4]$ satisfying $f(x) f(4-x) = 1$.

The value of the definite integral $\int_0^4 \frac{1}{1+f(x)} dx$ equals-

- (A) 0 (B) 1 (C) 2 (D) 4

DI0020

12. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)} dx$ is

[JEE 2011, 3 (-1)]

- (A) $\frac{1}{4} \ln \frac{3}{2}$ (B) $\frac{1}{2} \ln \frac{3}{2}$ (C) $\ln \frac{3}{2}$ (D) $\frac{1}{6} \ln \frac{3}{2}$

DI0156

13. The value of the defined integral $\int_0^{\pi/2} (\sin x + \cos x) \cdot \sqrt{\frac{e^x}{\sin x}} dx$ equals

- (A) $2\sqrt{e^{\pi/2}}$ (B) $\sqrt{e^{\pi/2}}$ (C) $2\sqrt{e^{\pi/2}} \cdot \cos 1$ (D) $\frac{1}{2} e^{\pi/4}$

DI0008

- 14.** Let f be a positive function. Let $I_1 = \int_{1-k}^k x f(x(1-x)) dx$; $I_2 = \int_{1-k}^k f(x(1-x)) dx$, where $2k - 1 > 0$.

Then $\frac{I_2}{I_1}$ is -

- (A) k (B) $1/2$ (C) 1 (D) 2

DI0017

- 15.** Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x} \right)^2 dx$ and $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x} \right)^2 dx$. Find the value of $\frac{v}{u}$.

- (A) 1 (B) 2 (C) 3 (D) 4

DI0062

- 16.** $\int_0^1 \frac{\tan^{-1} x}{x} dx =$

- (A) $\int_0^{\pi/4} \frac{\sin x}{x} dx$ (B) $\int_0^{\pi/2} \frac{x}{\sin x} dx$ (C) $\frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx$ (D) $\frac{1}{2} \int_0^{\pi/4} \frac{x}{\sin x} dx$

DI0010

17. The value of the definite integral $\int_1^e ((x+1)e^x \cdot \ln x) dx$ is -

- (A) e (B) e^{e+1} (C) $e^e(e-1)$ (D) $e^e(e-1)+e$

DI0012

- 18.** If $g(x)$ is the inverse of $f(x)$ and $f(x)$ has domain $x \in [1, 5]$, where $f(1) = 2$ and $f(5) = 10$ then the values

of $\int_1^5 f(x) dx + \int_2^{10} g(y) dy$ equals -

- (A) 48 (B) 64 (C) 71 (D) 52

DI0014

- 19.** Let f be a continuous function satisfying $f'(x) = \begin{cases} 1 & \text{for } 0 < x \leq 1 \\ x & \text{for } x > 1 \end{cases}$ and $f(0) = 0$ then $f(x)$ can be defined as

- $$(A) f(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ 1 - e^x & \text{if } x > 0 \end{cases}$$

- $$(B) f(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ e^x - 1 & \text{if } x > 0 \end{cases}$$

- $$(C) f(x) = \begin{cases} x & \text{if } x < 0 \\ e^x & \text{if } x > 0 \end{cases}$$

- $$(D) f(x) = \begin{cases} x & \text{if } x \leq 0 \\ e^x - 1 & \text{if } x > 0 \end{cases}$$

DI0034

20. $\int_0^{\infty} f\left(x + \frac{1}{x}\right) \cdot \frac{\ln x}{x} dx$

- (A) is equal to zero (B) is equal to one (C) is equal to $\frac{1}{2}$ (D) can not be evaluated

DI0018

21. The value of the definite integral $\int_0^{\pi/2} \sin |2x - \alpha| dx$ where $\alpha \in [0, \pi]$

- (A) 1 (B) $\cos \alpha$ (C) $\frac{1 + \cos \alpha}{2}$ (D) $\frac{1 - \cos \alpha}{2}$

DI0035

22. If $\int_0^x f(t) dt = x + \int_x^1 t^2 \cdot f(t) dt + \frac{\pi}{4} - 1$, then the value of the integral $\int_{-1}^1 f(x) dx$ is equal to

- (A) 0 (B) $\pi/4$ (C) $\pi/2$ (D) π

DI0022

23. Let $u = \int_0^1 \frac{\ln(x+1)}{x^2+1} dx$ and $v = \int_0^{\pi/2} \ln(\sin 2x) dx$ then -

- (A) $u = 4v$ (B) $4u + v = 0$ (C) $u + 4v = 0$ (D) $2u + v = 0$

DI0026

24. Let $I_1 = \int_0^{\pi/2} e^{-x^2} \sin(x) dx$; $I_2 = \int_0^{\pi/2} e^{-x^2} dx$; $I_3 = \int_0^{\pi/2} e^{-x^2} (1+x) dx$

and consider the statements

- I** $I_1 < I_2$ **II** $I_2 < I_3$ **III** $I_1 = I_3$

Which of the following is(are) true ?

- (A) I only (B) II only
(C) Neither I nor II nor III (D) Both I and II

DI0028

25. Variable x and y are related by equation $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$. The value of $\frac{d^2y}{dx^2}$ is equal to

- (A) $\frac{y}{\sqrt{1+y^2}}$ (B) y (C) $\frac{2y}{\sqrt{1+y^2}}$ (D) $4y$

DI0007

26. $\lim_{x \rightarrow \infty} \left(x^3 \int_{-1/x}^{1/x} \frac{\ln(1+t^2)}{1+e^t} dt \right)$ equals

- (A) $1/3$ (B) $2/3$ (C) 1 (D) 0

DI0023

27. The true solution set of the inequality, $\sqrt{5x-6-x^2} + \left(\frac{\pi}{2} \int_0^x dz \right) > x \int_0^\pi \sin^2 x dx$ is :

- (A) $\mathbb{R}$ (B) $(1,6)$ (C) $(-6,1)$ (D) $(2,3)$

DI0025

28. If $f(x) = x \sin x^2$; $g(x) = x \cos x^2$ for $x \in [-1, 2]$

$A = \int_{-1}^2 f(x) dx$; $B = \int_{-1}^2 g(x) dx$, then

- (A) $A > 0$; $B < 0$ (B) $A < 0$; $B > 0$ (C) $A > 0$; $B > 0$ (D) $A < 0$; $B < 0$

DI0030

29. $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{4n}}$ is equal to -

- (A) 2 (B) 4 (C) $2(\sqrt{2}-1)$ (D) $2\sqrt{2}-1$

DI0029

30. If $S_n = \left[\frac{1}{1+\sqrt{n}} + \frac{1}{2+\sqrt{2n}} + \dots + \frac{1}{n+\sqrt{n^2}} \right]$, then value of $\lim_{n \rightarrow \infty} S_n$ is

- (A) $\ln 2$ (B) $\ln 3$ (C) $\ln 4$ (D) $\ln 6$

EXERCISE (O-2)

Multiple Correct Answer Type

- Let $f(\alpha) = \int_0^\alpha \frac{(x-\alpha)}{\ln \alpha} dx$ ($\alpha > 0$), then -

(A) $f(e) = \frac{e}{2}$
(B) $f(e) = -\frac{e^2}{2}$
(C) $f'(e) = -\frac{e}{2}$
(D) $f(\alpha) = -\frac{\alpha^2}{2} \log_\alpha e$
- Suppose f is continuous and satisfies $f(x) + f(-x) = x^2$ then the integral $\int_{-1}^1 f(x) dx$ has the value less than

(A) $\frac{2}{3}$
(B) $\frac{1}{3}$
(C) $\frac{4}{3}$
(D) $\frac{5}{4}$
- Let $f(x) = \begin{cases} x+1, & 0 \leq x \leq 1 \\ 2x^2 - 6x + 6, & 1 < x \leq 2 \end{cases}$ and $g(t) = \int_{t-1}^t f(x) dx$ for $t \in [1, 2]$
 Which of the following hold(s) good ?

(A) $f(x)$ is continuous and differentiable in $[0, 2]$
(B) $g'(t)$ vanishes for $t = 3/2$ and 2
(C) $g(t)$ is maximum at $t = 3/2$
(D) $g(t)$ is minimum at $t = 1$
- Which of the following pair(s) of functions are the primitive of one and the same function ?

(A) $f(x) = \ln ax$; $g(x) = \ln x$
(B) $f(x) = 2 \sin^2 x$; $g(x) = -\cos 2x$
(C) $f(x) = (e^x + e^{-x})^2$; $g(x) = (e^x - e^{-x})^2$
(D) $f(x) = \frac{e^x (e^x + e^{-x})}{2}$; $g(x) = \frac{e^x (e^x - e^{-x})}{2}$
- Which one of the following functions is continuous on $(0, \pi)$?

(A) $f(x) = \cot x$

(B) $g(x) = \int_0^x t \sin \frac{1}{t} dt$

(C) $h(x) = \begin{cases} 1 & 0 < x \leq \frac{3\pi}{4} \\ 2 \sin \frac{2}{9} x & \frac{3\pi}{4} < x < \pi \end{cases}$

(D) $l(x) = \begin{cases} x \sin x, & 0 < x \leq \frac{\pi}{2} \\ \frac{\pi}{2} \sin(x + \pi), & \frac{\pi}{2} < x < \pi \end{cases}$

6. The true set of values of 'a' for which the inequality $\int_a^0 (3^{-2x} - 2.3^{-x})dx \geq 0$ is true are -

(A) $[0,1]$ (B) $(-\infty,-1]$ (C) $[0,\infty)$ (D) $(-\infty,-1] \cup [0,\infty)$

DI0045

7. $\int_0^{\infty} \frac{x}{(1+x)(1+x^2)} dx =$

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) is same as $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)}$

(D) cannot be evaluated

DI0047

8. Let $u = \int_0^{\infty} \frac{dx}{x^4 + 7x^2 + 1}$ & $v = \int_0^{\infty} \frac{x^2 dx}{x^4 + 7x^2 + 1}$ then -

(A) $v > u$ (B) $6v = \pi$ (C) $3u + 2v = 5\pi/6$ (D) $u + v = \pi/3$

DI0048

9. Let $F(x) = \int_{\sin x}^{\cos x} e^{(1+\arcsin t)^2} dt$ on $\left[0, \frac{\pi}{2}\right]$ then

(A) $F''(c) = 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$

(B) $F''(c) = 0$ for some $c \in \left(0, \frac{\pi}{2}\right)$

(C) $F'(c) \neq 0$ for all $c \in \left(0, \frac{\pi}{2}\right)$

(D) $F(c) = 0$ for some $c \in \left(0, \frac{\pi}{2}\right)$

DI0053

- 10.** Let $f: \left[0, \frac{\pi}{2}\right] \rightarrow [0, 1]$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 1$, then

$$(A) \ f'(\alpha) = \sqrt{1-f^2(\alpha)} \text{ for all } \alpha \in \left(0, \frac{\pi}{2}\right)$$

(B) $f'(\alpha) = \frac{2}{\pi}$ for all $\alpha \in \left(0, \frac{\pi}{2}\right)$

(C) $f(\alpha) f'(\alpha) = \frac{1}{\pi}$ for atleast one $\alpha \in \left(0, \frac{\pi}{2}\right)$

(D) $f'(\alpha) = \frac{8\alpha}{\pi^2}$ for atleast one $\alpha \in \left(0, \frac{\pi}{2}\right)$

DI0054

11. Which of the following is/are true -

(A) $\int_0^{\infty} e^{-x^2} dx = \int_0^{\infty} \frac{e^{-x}}{\sqrt{x}} dx$

(B) $\int_0^{\infty} \frac{\sin x}{x} dx = \int_0^{\infty} \frac{\sin 2x}{x} dx$

(C) $\int_0^{\infty} \frac{\sin x}{x} dx = \int_0^{\infty} \frac{\sin^2 x}{x^2} dx$

(D) $\int_0^{\pi} x \ln \sin x dx = -\frac{\pi^2}{2} \ln 2$

DI0227

12. $f(x) = -4 \int_{\left(\frac{3\pi}{2} - \alpha\right)}^x \sin 4\theta d\theta$, where α is acute angle, then the factors of $f(x)$ is/are -

DI0228

(A) $\cos x + \cos \alpha$

(B) $\cos x - \cos \alpha$

(C) $\cos x - \sin \alpha$

(D) $(\cos x + \sin \alpha)$

13. If $f(x) = \int_1^x \frac{\ell n t}{1+t} dt$, then -

DI0229

(A) $f\left(\frac{1}{x}\right) = -\int_1^x \frac{\ell n t}{t(1+t)} dt$

(B) $f\left(\frac{1}{x}\right) = \int_1^x \frac{\ell n t}{t(1+t)} dt$

(C) $f(x) + f\left(\frac{1}{x}\right) = \frac{1}{2} \ell n^2(x)$

(D) $f(x) = -f\left(\frac{1}{x}\right)$

14. Let $S_n = \sum_{k=1}^n \frac{k^2 + n^2}{n^3}$ and $T_n = \sum_{k=0}^{n-1} \frac{k^2 + n^2}{n^3}$ for $n = 1, 2, 3, \dots$. Then -

(A) $S_n < \frac{4}{3}$

(B) $T_n > \frac{4}{3}$

(C) $S_n > \frac{4}{3}$

(D) $T_n < \frac{4}{3}$

DI0050

15. Which of the following statement(s) is/are **TRUE**?

(A) $\int_0^1 \ell n x dx = -1$

(B) $\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right) = 1 + 2 \ln 2$.

(C) Let f be a continuous and non-negative function defined on $[a, b]$.

If $\int_a^b f(x) dx = 0$ then $f(x) = 0 \forall x \in [a, b]$

(D) Let f be a continuous function defined on $[a, b]$ such that $\int_a^b f(x) dx = 0$, then there exists atleast one

$c \in (a, b)$ for which $f(c) = 0$.

DI0051

16. If $\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\frac{\sqrt{a^2+b^2}}{2}} \frac{x \cdot dx}{\sqrt{(x^2 - a^2)(b^2 - x^2)}}$ is equal to k, then which of the following option(s) is/are **True**?

DI0230

(A) $\sin 2k = \frac{1}{2}$ (B) $\cos 2k = \frac{\sqrt{3}}{2}$ (C) $\tan 4k = \sqrt{3}$ (D) $\tan 2k = \frac{1}{\sqrt{3}}$

- 17.** Let $\beta(n) = \int_0^{n\pi} \sqrt{1 - \sin t} \, dt$. Then the value of $\beta(2) - \beta(1)$ is divisible by

DI0231

(A) 2 (B) 4 (C) 3 (D) 5

18. If $\int_0^1 \frac{2-x^2}{(1+x)\sqrt{1-x^2}} dx$ is equal to k, then which of the following option(s) is/are **True** ?

DI0232

(A) $\sin k = 1$ (B) $\cos k = 0$ (C) $\sin \frac{k}{2} = \frac{1}{\sqrt{2}}$ (D) $\cos \frac{k}{2} = \frac{1}{\sqrt{2}}$

- 19.** Let $f(x) = \begin{cases} -1 & \text{if } -2 \leq x \leq 0 \\ |x-1| & \text{if } 0 < x \leq 2 \end{cases}$ and $g(x) = \int_{-2}^x f(t) dt$. If ℓ denotes number of points of discontinuity

and m denotes number of points of non-differentiability of $g(x)$ in $(-2, 2)$, then

DI0233

(A) $\ell = 0$ (B) $m = 1$ (C) $\ell = 1$ (D) $m = 2$

- 20.** If $U_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx$, then

DI0234

(A) $U_1, U_2, \dots, U_n$ forms A.P.

(B) $U_1, U_2, \dots, U_n$ forms H.P.

$$(C) U_n = \frac{n\pi}{2}$$

(D) $U_{10} = 5\pi$

- 21.** Determine a pair of number a and b for which $\int_0^1 \frac{ax+b}{(x^2+3x+2)^2} dx = \frac{5}{2}$, then

DI0128

(A) $a = 15$ (B) $a = \frac{45}{2}$ (C) $b = 15$ (D) $b = \frac{45}{2}$

22. If $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \left[k \int_k^{k+1} \sqrt{(x-k)(k+1-x)} dx \right]$ is equal to k , then which of the following option(s) is/are True?

- (A) $\tan 4k = 1$ (B) $\cot 4k = \sqrt{3}$ (C) $\sin 4k = \frac{1}{2}$ (D) $\cos 4k = \frac{\sqrt{3}}{2}$ **DI0235**

23. If $g(x) = \int_0^x \cos 4t dt$, then $g(x + \pi)$ equals :

- (A) $g(x) \cdot g(\pi)$ (B) $\frac{g(x)}{g(\pi)}$ (C) $g(x) + g(\pi)$ (D) $g(x) - g(\pi)$

DI0141

24. Let S be the area of the region enclosed by $y = e^{-x^2}$, $y = 0$, $x = 0$, and $x = 1$. Then -

- (A) $S \geq \frac{1}{e}$ (B) $S \geq 1 - \frac{1}{e}$
 (C) $S \leq \frac{1}{4} \left(1 + \frac{1}{\sqrt{e}} \right)$ (D) $S \leq \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{e}} \left(1 - \frac{1}{\sqrt{2}} \right)$

DI0157

25. If $f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$ for all $x \in (0, \infty)$, then -

- (A) f has a local maximum at $x = 2$
 (B) f is decreasing on $(2, 3)$
 (C) there exists some $c \in (0, \infty)$ such that $f''(c) = 0$
 (D) f has a local minimum at $x = 3$

DI0160

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Let the function f satisfies

$$f(x) \cdot f'(-x) = f(-x) \cdot f'(x) \text{ for all } x \text{ and } f(0) = 3.$$

1. The value of $f(x) \cdot f(-x)$ for all x , is

(A) 4 (B) 9 (C) 12 (D) 16

DI0041

2. $\int_{-51}^{51} \frac{dx}{3 + f(x)}$ has the value equal to

(A) 17 (B) 34 (C) 102 (D) 0

DI0041

3. Number of roots of $f(x) = 0$ in $[-2, 2]$ is

(A) 0 (B) 1 (C) 2 (D) 4

DI0041

Paragraph for Question 4 and 5

Let $P(x)$ be a polynomial satisfying $\lim_{x \rightarrow \infty} \frac{xP(x)}{2016 + x^6} = 1$

where $P(0) = 1$, $P(2) = 9$, $P(3) = 28$, $P(4) = 65$ and $P(6) = 217$, then

4. $\int_{-3}^9 (P(x) - x^3) dx$ is -

(A) 0 (B) 12 (C) $\frac{P(5)}{8}$ (D) $\frac{-P(-1)}{35}$

5. If $\int \frac{(x-3)}{(P(x) - x^3 - 1)} dx = \frac{1}{A} \ln \left| \frac{x-6}{x} \right| + \frac{1}{B} \ln \left| \frac{x-2}{x-4} \right| + C$ (where C is an integration constant), then -

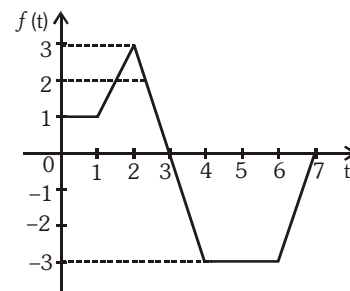
(A) $A - 3B = 0$ (B) $B^2 - 5A = 16$ (C) $A = \frac{1}{48}$ (D) $B = 16$

Paragraph for Question 6 to 8

Let $g(x) = \int_0^x f(t) dt$, where f is a function

whose graph is show adjacently.

On the basis of above information, answer the following questions :



6. Maximum value of $g(x)$ in $x \in [0, 7]$ is -
7. Value of x at which $g(x)$ becomes zero, is -
8. The value of $x \in \mathbb{N}$ in $[0, 7]$ for which $g(x)$ is negative is -

Matrix Match Type

9. List-I

List-II

(P) $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \left(1 + \frac{3^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$ (1) $3 - \ln 4$

DI0094

(Q) $\lim_{n \rightarrow \infty} \left[\frac{n!}{n^n} \right]^{1/n}$ (2) $\tan^{-1} 2$

DI0133

(R) $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{1}{n+1} + \frac{2}{n+2} + \dots + \frac{3n}{4n} \right]$ (3) $2 e^{(1/2)(\pi - 4)}$

DI0095

(S) $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \frac{n}{n^2 + 3^2} + \dots + \frac{1}{5n} \right)$ is equal to (4) $\frac{1}{e}$

DI0236

(A) (P) $\rightarrow$ (3), (Q) $\rightarrow$ (4), (R) $\rightarrow$ (1), (S) $\rightarrow$ (2)

(B) (P) $\rightarrow$ (4), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (2), (S) $\rightarrow$ (1)

(C) (P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (4), (S) $\rightarrow$ (3)

(D) (P) $\rightarrow$ (2), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (1), (S) $\rightarrow$ (4)

10. Column-I

Column-II

(A) $\lim_{n \rightarrow \infty} n^2 \int_{-1/n}^{1/n} (2010 \sin x + 2012 \cos x) |x| dx$

(P) 13.5

DI0088

(B) Evaluate : $\lim_{x \rightarrow +\infty} \frac{d}{dx} \int_{2 \sin^{-1} \frac{1}{x}}^{3\sqrt{x}} \frac{3t^4 + 1}{(t-3)(t^2+3)} dt$

(Q) $\frac{1}{2}$

DI0090

(C) $\lim_{n \rightarrow \infty} \frac{\int_{1/(n+1)}^{1/n} \tan^{-1}(nx) dx}{\int_{1/(n+1)}^{1/n} \sin^{-1}(nx) dx}$ is equal to

(R) 12

DI0237

(D) The value of $\frac{\frac{d}{dx} \int_0^{x^3} \sqrt{\cos t} dt}{1 - \sqrt{\cos x}}$

(S) 2012

DI0238

EXERCISE (O-4)

Numerical Grid Type

1. Let $I_n = \int_{-n}^n (\{x+1\} \cdot \{x^2+2\} + \{x^2+2\} \{x^3+4\}) dx$, where $\{.\}$ denotes the fractional part of x . Find I_1 .

DI0065

2. Suppose f is continuous, $f(0) = 0$, $f(1) = 1$, $f'(x) > 0$ and $\int_0^1 f(x) dx = \frac{1}{3}$. Find the value of the

definite integral $\int_0^1 f^{-1}(y) dy$.

DI0111

3. Let $U_{10} = \int_0^{\frac{\pi}{2}} x \sin^{10} x dx$, then find the value of $\left(\frac{100U_{10} - 1}{U_8} \right)$.

DI0087

4. Let $y = f(x)$ be a quadratic function with $f'(2) = 1$. Find the value of the integral $\int_{2-\pi}^{2+\pi} f(x) \cdot \sin\left(\frac{x-2}{2}\right) dx$.

DI0117

5. If a_1, a_2 and a_3 are the three values of a which satisfy the equation

$$\int_0^{\pi/2} (\sin x + a \cos x)^3 dx - \frac{4a}{\pi-2} \int_0^{\pi/2} x \cos x dx = 2 \text{ then find the value of } 1000(a_1^2 + a_2^2 + a_3^2).$$

DI0074

6. $\int_1^2 \frac{(x^2-1)dx}{x^3 \cdot \sqrt{2x^4-2x^2+1}} = \frac{u}{v}$ where u and v are in their lowest form. Find the value of $\frac{(1000)u}{v}$.

DI0071

7. If $\int_0^{\pi} \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} dx$ has the value equal to $\left(\frac{\pi}{k} + \sqrt{w} \right)$ where

k and w are positive integers, find the value of $(k^2 + w^2)$.

DI0064

8. Let $f(x)$ be a function defined on R such that $f'(x) = f'(3-x) \forall x \in [0,3]$ with $f(0) = -32$ and $f(3) = 46$. Then find the value of $\int_0^3 f(x)dx$.

DI0086

9. A continuous real function f satisfies $f(2x) = 3f(x) \forall x \in R$

If $\int_0^1 f(x)dx = 1$, then compute the value of definite integral $\int_1^2 f(x)dx$

DI0078

10. If $f(x) = x + \sin x$ and I denotes the value of integral $\int_{\pi}^{2\pi} (f^{-1}(x) + \sin x)dx$ then the value of $\left[\frac{2I}{3} \right]$

(where $[.]$ denotes greatest integer function)

DI0126

EXERCISE (JM)

1. **Statement-I :** The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}}$ is equal to $\frac{\pi}{6}$. [JEE-MAIN-2013]

Statement-II : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$.

- (1) Statement-I is true, Statement-II is true; Statement-II is a **correct** explanation for Statement-I.
 (2) Statement-I is true, Statement-II is true; Statement-II is **not** a correct explanation for Statement-I.
 (3) Statement-I is true, Statement-II is false.
 (4) Statement-I is false, Statement-II is true.

DI0142

2. The intercepts on x-axis made by tangents to the curve, $y = \int_0^x |t| dt$, $x \in \mathbb{R}$, which are parallel to the line $y = 2x$, are equal to [JEE-MAIN 2013]

- (1) ± 1 (2) ± 2 (3) ± 3 (4) ± 4

DI0143

3. The integral $\int_0^{\pi} \sqrt{1 + 4 \sin^2 \frac{x}{2}} - 4 \sin \frac{x}{2} dx$ equals :

[JEE-MAIN-2014]

- (1) $\pi - 4$ (2) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$ (3) $4\sqrt{3} - 4$ (4) $4\sqrt{3} - 4 - \frac{\pi}{3}$

DI0144

4. The integral $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)} dx$ is equal to :

[JEE-MAIN-2015]

- (1) 1 (2) 6 (3) 2 (4) 4

DI0145

5. The integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1 + \cos x}$ is equal to :-

[JEE-MAIN-2017]

- (1) -1 (2) -2 (3) 2 (4) 4

DI0146

6. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+2^x} dx$ is : [JEE-MAIN-2018]

- (1) $\frac{\pi}{2}$ (2) 4π (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{8}$

DI0147

7. The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$, where $[t]$ denotes the greatest integer less than or equal to t , is : [JEE-MAIN 2019]

- (1) $\frac{1}{12}(7\pi+5)$ (2) $\frac{3}{10}(4\pi-3)$ (3) $\frac{1}{12}(7\pi-5)$ (4) $\frac{3}{20}(4\pi-3)$

DI0148

8. If $\int_0^x f(t) dt = x^2 + \int_x^1 t^2 f(t) dt$, then $f(1/2)$ is : [JEE-MAIN 2019]

- (1) $\frac{6}{25}$ (2) $\frac{24}{25}$ (3) $\frac{18}{25}$ (4) $\frac{4}{5}$

DI0149

9. Let $I = \int_a^b (x^4 - 2x^2) dx$. If I is minimum then the ordered pair (a, b) is : [JEE-MAIN 2019]

- (1) $(-\sqrt{2}, 0)$ (2) $(-\sqrt{2}, \sqrt{2})$ (3) $(0, \sqrt{2})$ (4) $(\sqrt{2}, -\sqrt{2})$

DI0150

10. $\lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \dots + \frac{1}{5n} \right)$ is equal to : [JEE-MAIN 2019]

- (1) $\frac{\pi}{4}$ (2) $\tan^{-1}(2)$ (3) $\tan^{-1}(3)$ (4) $\frac{\pi}{2}$

DI0151

11. The value of α for which $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$, is : [JEE-MAIN 2020]

- (1) $\log_e \left(\frac{3}{2} \right)$ (2) $\log_e \left(\frac{4}{3} \right)$ (3) $\log_e 2$ (4) $\log_e \sqrt{2}$

DI0152

12. If $f(a + b + 1 - x) = f(x)$, for all x , where a and b are fixed positive real numbers, then

$$\frac{1}{a+b} \int_a^b x(f(x) + f(x+1)) dx \text{ is equal to :}$$

[JEE-MAIN 2020]

(1) $\int_{a+1}^{b+1} f(x) dx$

(2) $\int_{a+1}^{b+1} f(x+1) dx$

(3) $\int_{a-1}^{b-1} f(x+1) dx$

(4) $\int_{a-1}^{b-1} f(x) dx$

DI0153

13. If $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$, then :

[JEE-MAIN 2020]

(1) $\frac{1}{9} < I^2 < \frac{1}{8}$

(2) $\frac{1}{16} < I^2 < \frac{1}{9}$

(3) $\frac{1}{6} < I^2 < \frac{1}{2}$

(4) $\frac{1}{8} < I^2 < \frac{1}{4}$

DI0154

14. The value of $\int_0^{2\pi} \frac{x \sin^8 x}{\sin^8 x + \cos^8 x} dx$ is equal to :

[JEE-MAIN 2020]

(1) 2π

(2) 4π

(3) $2\pi^2$

(4) π^2

DI0155

15. Let a be a positive real number such that $\int_0^a e^{x-[x]} dx = 10e - 9$ where $[x]$ is the greatest integer less than or equal to x . Then a is equal to :

[JEE-MAIN 2021]

(1) $10 - \log_e(1+e)$

(2) $10 + \log_e 2$

(3) $10 + \log_e 3$

(4) $10 + \log_e(1+e)$

DI0219

16. Let $f: [0, \infty) \rightarrow [0, \infty)$ be defined as $f(x) = \int_0^x [y] dy$

[JEE-MAIN 2021]

where $[x]$ is the greatest integer less than or equal to x . Which of the following is true ?

(1) f is continuous at every point in $[0, \infty)$ and differentiable except at the integer points.

(2) f is both continuous and differentiable except at the integer points in $[0, \infty)$.

(3) f is continuous everywhere except at the integer points in $[0, \infty)$.

(4) f is differentiable at every point in $[0, \infty)$.

DI0220

17. If $f(x) = \begin{cases} \int_0^x (5 + |1-t|) dt & ; x > 2 \\ 5x + 1 & ; x \leq 2 \end{cases}$, then [JEE-MAIN 2021]

- (1) $f(x)$ is not continuous at $x = 2$
 (2) $f(x)$ is everywhere differentiable
 (3) $f(x)$ is continuous but not differentiable at $x = 2$
 (4) $f(x)$ is not differentiable at $x = 1$

DI0221

18. The value of the definite integral $\int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{x \cos x})(\sin^4 x + \cos^4 x)}$ is equal to : [JEE-MAIN 2021]

- (1) $-\frac{\pi}{2}$ (2) $\frac{\pi}{2\sqrt{2}}$ (3) $-\frac{\pi}{4}$ (4) $\frac{\pi}{\sqrt{2}}$ DI0222

19. The value of the integral $\int_{-\pi/2}^{\pi/2} \frac{dx}{(1 + e^x)(\sin^6 x + \cos^6 x)}$ is equal to [JEE-MAIN 2022]

- (A) 2π (B) 0 (C) π (D) $\frac{\pi}{2}$ DI0223

20. The value of $\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1 + \cos^2 x)(e^{\cos x} + e^{-\cos x})} dx$ is equal to [JEE-MAIN 2022]

- (A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$ DI0224

21. Let $f(x) = \max\{|x + 1|, |x + 2|, \dots, |x + 5|\}$. Then $\int_{-6}^0 f(x) dx$ is equal to _____.

[JEE-MAIN 2022]

DI0225

22. The value of the integral $\int_{-\pi/4}^{\pi/4} \frac{x + \frac{\pi}{4}}{2 - \cos 2x} dx$ is : [JEE-MAIN 2022]

- (1) $\frac{\pi^2}{6}$ (2) $\frac{\pi^2}{12\sqrt{3}}$ (3) $\frac{\pi^2}{3\sqrt{3}}$ (4) $\frac{\pi^2}{6\sqrt{3}}$ DI0226

EXERCISE (JA)

1. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$. $\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1}[(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$

Then $a =$

[JEE(Advanced) 2013, 3, (-1)]

- (A) 5 (B) 7 (C) $-\frac{15}{2}$ (D) $-\frac{17}{2}$

DI0161

2. Let $f : [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < a \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b \\ \int_a^b f(t) dt & \text{if } x > b \end{cases}$$

Then

[JEE(Advanced)-2014, 3]

- (A) $g(x)$ is continuous but not differentiable at a
 (B) $g(x)$ is differentiable on $\mathbb{R}$
 (C) $g(x)$ is continuous but not differentiable at b
 (D) $g(x)$ is continuous and differentiable at either a or b but not both.

DI0162

3. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx$ is

[JEE(Advanced)-2014, 3]

DI0163

4. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to -

[JEE(Advanced)-2014, 3(-1)]

(A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$

(B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$

(C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$

(D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

DI0164

5. Let $f : [0, 2] \rightarrow \mathbb{R}$ be a function which is continuous on $[0, 2]$ and is differentiable on $(0, 2)$ with $f(0) = 1$. Let $F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$ for all $x \in (0, 2)$, then $F(2)$ equals -
- (A) $e^2 - 1$ (B) $e^4 - 1$ (C) $e - 1$ (D) e^4

[JEE(Advanced)-2014, 3(-1)]

DI0165

Paragraph For Questions 6 and 7

Given that for each $a \in (0, 1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a} (1-t)^{a-1} dt$ exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0, 1)$.

6. The value of $g\left(\frac{1}{2}\right)$ is -
- (A) π (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

[JEE(Advanced)-2014, 3(-1)]

DI0166

7. The value of $g'\left(\frac{1}{2}\right)$ is-
- (A) $\frac{\pi}{2}$ (B) π (C) $-\frac{\pi}{2}$ (D) 0

[JEE(Advanced)-2014, 3(-1)]

DI0166

8. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = \int_{\frac{1}{x}}^x e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$. Then
- (A) $f(x)$ is monotonically increasing on $[1, \infty)$
 (B) $f(x)$ is monotonically decreasing on $[0, 1)$
 (C) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$
 (D) $f(2^x)$ is an odd function of x on $\mathbb{R}$

[JEE(Advanced)-2014, 3]

DI0167

9. List-I

List-II

- P.** The number of polynomials $f(x)$ with non-negative integer coefficients of degree < 2 , satisfying

$f(0) = 0$ and $\int_0^1 f(x)dx = 1$, is

DI0168

- Q.** The number of points in the interval $[-\sqrt{13}, \sqrt{13}]$ at which $f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is

DI0169

- R.** $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals

DI0170

- S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x}\right) dx\right)}$ equals

DI0171

Codes :

	P	Q	R	S
(A)	3	2	4	1
(B)	2	3	4	1
(C)	3	2	1	4
(D)	2	3	1	4

[JEE(Advanced)-2014, 3(-1)]

- 10.** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x] & , \quad x \leq 2 \\ 0 & , \quad x > 2 \end{cases}$,

where $[x]$ is the greatest integer less than or equal to x . If $I = \int_{-1}^2 \frac{x f(x^2)}{2 + f(x+1)} dx$, then the value of

$(4I - 1)$ is

[JEE 2015, 4M, -0M]

DI0172

11. If $\alpha = \int_0^1 \left(e^{9x+3\tan^{-1}x} \right) \left(\frac{12+9x^2}{1+x^2} \right) dx$, where $\tan^{-1}x$ takes only principal values, then the value of

$$\left(\log_e \left| 1 + \alpha \right| - \frac{3\pi}{4} \right) \text{ is}$$

[JEE 2015, 4M, -0M]

DI0173

12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose

that $F(x) = \int_{-1}^x f(t) dt$ for all $x \in [-1, 2]$ and $G(x) = \int_{-1}^x |f(f(t))| dt$ for all $x \in [-1, 2]$. If $\lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}$,

then the value of $f\left(\frac{1}{2}\right)$ is

[JEE 2015, 4M, -0M]

DI0174

13. The option(s) with the values of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L ?$$

[JEE 2015, 4M, -0M]

(A) $a = 2, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

(B) $a = 2, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$

(C) $a = 4, L = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$

(D) $a = 4, L = \frac{e^{4\pi} + 1}{e^{\pi} + 1}$

DI0175

14. Let $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression(s) is(are)

(A) $\int_0^{\pi/4} x f(x) dx = \frac{1}{12}$

(B) $\int_0^{\pi/4} f(x) dx = 0$

[JEE 2015, 4M, -0M]

(C) $\int_0^{\pi/4} x f(x) dx = \frac{1}{6}$

(D) $\int_0^{\pi/4} f(x) dx = 1$

DI0176

15. Let $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \leq \int_{1/2}^1 f(x) dx \leq M$, then the possible values of m and M are [JEE 2015, 4M, -0M]

- (A) $m = 13, M = 24$ (B) $m = \frac{1}{4}, M = \frac{1}{2}$
 (C) $m = -11, M = 0$ (D) $m = 1, M = 12$

DI0177

Paragraph For Questions 16 and 17

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. Suppose that $F(1) = 0, F(3) = -4, F'(x) < 0$ for all $x \in (1/2, 3)$. Let $f(x) = xF(x)$ for all $x \in \mathbb{R}$.

16. The correct statement(s) is(are) [JEE 2015, 4M, -0M]
 (A) $f'(1) < 0$ (B) $f(2) < 0$
 (C) $f'(x) \neq 0$ for any $x \in (1, 3)$ (D) $f'(x) = 0$ for some $x \in (1, 3)$

DI0178

17. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 x^3 F''(x) dx = 40$, then the correct expression(s) is(are)

- (A) $9f'(3) + f'(1) - 32 = 0$ (B) $\int_1^3 f(x) dx = 12$ [JEE 2015, 4M, -0M]
 (C) $9f'(3) - f'(1) + 32 = 0$ (D) $\int_1^3 f(x) dx = -12$

DI0178

18. The value of $\int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1 + e^x} dx$ is equal to [JEE(Advanced)-2016, 3(-1)]

- (A) $\frac{\pi^2}{4} - 2$ (B) $\frac{\pi^2}{4} + 2$ (C) $\pi^2 - e^{\frac{\pi}{2}}$ (D) $\pi^2 + e^{\frac{\pi}{2}}$

DI0179

19. The total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^2}{1+t^4} dt = 2x - 1$ is

[JEE(Advanced)-2016, 3(0)]

DI0180

20. Let $f(x) = \lim_{n \rightarrow \infty} \frac{n^n (x+n) \left(x + \frac{n}{2}\right) \dots \left(x + \frac{n}{n}\right)}{n! (x^2 + n^2) \left(x^2 + \frac{n^2}{4}\right) \dots \left(x^2 + \frac{n^2}{n^2}\right)}^{x/n}$, for all $x > 0$. Then

[JEE(Advanced)-2016, 4(-2)]

(A) $f\left(\frac{1}{2}\right) \geq f(1)$ (B) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$ (C) $f'(2) \leq 0$ (D) $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

DI0181

- 21.** Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$. If

$$g(x) = \int_x^{\frac{\pi}{2}} [f'(t) \operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt$$

for $x \in \left(0, \frac{\pi}{2}\right]$, then $\lim_{x \rightarrow 0} g(x) =$

[JEE(Advanced)-2017, 3]

DI0182

- 22.** If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then

[JEE(Advanced)-2017, 4]

(A) $I < \frac{49}{50}$ (B) $I < \log_e 99$ (C) $I > \frac{49}{50}$ (D) $I > \log_e 99$

DI0183

- 23.** If $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t) dt$, then

[JEE(Advanced)-2017, 4]

(A) $g'\left(\frac{\pi}{2}\right) = -2\pi$

(B) $g'\left(-\frac{\pi}{2}\right) = 2\pi$

(C) $g'\left(\frac{\pi}{2}\right) = 2\pi$

(D) $g'\left(-\frac{\pi}{2}\right) = -2\pi$

DI0184

24. Let $f : \mathbb{R} \rightarrow (0, 1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval $(0, 1)$? [JEE(Advanced)-2017]

(A) $e^x - \int_0^x f(t) \sin t dt$

(B) $x^9 - f(x)$

(C) $f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t dt$

(D) $x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t dt$

DI0185

25. For each positive integer n , let

$$y_n = \frac{1}{n} (n+1)(n+2)\dots(n+n)^{1/n}$$

For $x \in \mathbb{R}$, let $[x]$ be the greatest integer less than or equal to x . If $\lim_{n \rightarrow \infty} y_n = L$, then the value of $[L]$ is _____ [JEE(Advanced)-2018, 3(0)]

DI0186

26. The value of the integral

$$\int_0^{\frac{1}{2}} \frac{1 + \sqrt{3}}{\left((x+1)^2 (1-x)^6\right)^{\frac{1}{4}}} dx$$

is _____ .

[JEE(Advanced)-2018, 3(0)]

DI0187

27. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x-1)(x-2)(x-5)$. Define $F(x) = \int_0^x f(t) dt$, $x > 0$. Then which of the following options is/are correct ? [JEE(Advanced)-2019, 4(-1)]

(1) F has a local minimum at $x = 1$

(2) F has a local maximum at $x = 2$

(3) $F(x) \neq 0$ for all $x \in (0, 5)$

(4) F has two local maxima and one local minimum in $(0, \infty)$

DI0188

28. If $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)}$ then $27I^2$ equals _____

[JEE(Advanced)-2019, 3(0)]

DI0189

29. For $a \in \mathbb{R}$, $|a| > 1$, let $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) = 54$. Then the possible value(s)

of a is/are :

[JEE(Advanced)-2019, 4(-1)]

(1) 8

(2) -9

(3) -6

(4) 7

DI0190

30. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ equals

[JEE(Advanced)-2019, 3(0)]

DI0191

31. Which of the following inequalities is/are TRUE ?

[JEE(Advanced)-2020]

(A) $\int_0^1 x \cos x dx \geq \frac{3}{8}$ (B) $\int_0^1 x \sin x dx \geq \frac{3}{10}$ (C) $\int_0^1 x^2 \cos x dx \geq \frac{1}{2}$ (D) $\int_0^1 x^2 \sin x dx \geq \frac{2}{9}$

DI0208

32. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$.

If $F : [0, \pi] \rightarrow \mathbb{R}$ is defined by $F(x) = \int_0^x f(t) dt$, and if

[JEE(Advanced)-2020]

$$\int_0^{\pi} (f'(x) + F(x)) \cos x dx = 2,$$

then the value of $f(0)$ is _____

DI0209

33. Let $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be a continuous function such that $f(0) = 1$ and $\int_0^{\frac{\pi}{3}} f(t) dt = 0$

Then which of the following statements is (are) TRUE ?

[JEE(Advanced)-2021]

DI0210

(A) The equation $f(x) - 3 \cos 3x = 0$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(B) The equation $f(x) - 3 \sin 3x = -\frac{6}{\pi}$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(C) $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} = -1$

(D) $\lim_{x \rightarrow 0} \frac{\sin x \int_0^x f(t) dt}{x^2} = -1$

Question Stem for Questions Nos. 34 and 35

Question Stem

Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow \mathbb{R}$, $i = 1, 2$, and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow \mathbb{R}$ be functions such that

$$g_1(x) = 1, g_2(x) = |4x - \pi| \text{ and } f(x) = \sin^2 x, \text{ for all } x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$$

$$\text{Define } S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx, i = 1, 2$$

[JEE(Advanced)-2021]

34. The value of $\frac{16S_1}{\pi}$ is _____.

DI0211

35. The value of $\frac{48S_2}{\pi^2}$ is _____.

DI0212

Paragraph

Let $\psi_1 : [0, \infty) \rightarrow \mathbb{R}$, $\psi_2 : [0, \infty) \rightarrow \mathbb{R}$, $f : [0, \infty) \rightarrow \mathbb{R}$ and $g : [0, \infty) \rightarrow \mathbb{R}$ be functions such that

$$f(0) = g(0) = 0,$$

$$\psi_1(x) = e^{-x} + x, x \geq 0,$$

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2, x \geq 0,$$

$$f(x) = \int_{-x}^x (|t| - t^2) e^{-t^2} dt, x > 0$$

and

$$g(x) = \int_0^{x^2} \sqrt{t} e^{-t} dt, x > 0$$

[JEE(Advanced)-2021]

36. Which of the following statements is **TRUE** ?

(A) $f(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = \frac{1}{3}$

(B) For every $x > 1$, there exists an $\alpha \in (1, x)$ such that $\psi_1(x) = 1 + \alpha x$

(C) For every $x > 0$, there exists a $\beta \in (0, x)$ such that $\psi_2(x) = 2x(\psi_1(\beta) - 1)$

(D) f is an increasing function on the interval $\left[0, \frac{3}{2}\right]$

DI0213

37. Which of the following statements is **TRUE** ?

[JEE(Advanced)-2021]

(A) $\psi_1(x) \leq 1$, for all $x > 0$

(B) $\psi_2(x) \leq 0$, for all $x > 0$

(C) $f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$, for all $x \in \left(0, \frac{1}{2}\right)$

(D) $g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$, for all $x \in \left(0, \frac{1}{2}\right)$

DI0214

38. For any real number x , let $[x]$ denote the largest integer less than or equal to x . If $I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx$, then the value of $9I$ is ____.

[JEE(Advanced)-2021]

DI0215

39. Consider the equation $\int_1^e \frac{(\log_e x)^{1/2}}{x(a - (\log_e x)^{3/2})^2} dx = 1$, $a \in (-\infty, 0) \cup (1, \infty)$.

Which of the following statements is/are **TRUE** ?

[JEE(Advanced)-2022]

(A) **No** a satisfies the above equation

(B) An integer a satisfies the above equation

(C) An irrational number a satisfies the above equation

(D) More than one a satisfy the above equation

DI0216

40. The greatest integer less than or equal to

[JEE(Advanced)-2022]

$$\int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx$$

is ____.

DI0217

41. For positive integer n , define

$$f(n) = n + \frac{16 + 5n - 3n^2}{4n + 3n^2} + \frac{32 + n - 3n^2}{8n + 3n^2} + \frac{48 - 3n - 3n^2}{12n + 3n^2} + \dots + \frac{25n - 7n^2}{7n^2}.$$

Then, the value of $\lim_{n \rightarrow \infty} f(n)$ is equal to

[JEE(Advanced)-2022]

(A) $3 + \frac{4}{3} \log_e 7$

(B) $4 - \frac{3}{4} \log_e \left(\frac{7}{3}\right)$

(C) $4 - \frac{4}{3} \log_e \left(\frac{7}{3}\right)$

(D) $3 + \frac{3}{4} \log_e 7$

DI0218

ANSWER KEY

Do yourself - 1

1. 2 2. (C) 3. 5 4. $\frac{9}{2}$ 5. $\frac{31}{6}$
6. 2 7. (A) 8. $2(\sqrt{2}-1)$ 9. $\ln 2$ 10. $\frac{1}{2} \ln\left(\frac{e}{2}\right)$
11. $\frac{1}{6}$ 12. $\ln \frac{4}{3}$ 13. $\frac{1}{3}$ 14. $|b| - |a|$ 15. $2\sqrt{e}$
16. $e - \frac{2}{\ln 2}$ 17. $\frac{\pi}{4}$ 18. $\frac{1}{6}$ 19. $\frac{\pi-3}{16}$ 20. $\frac{2}{3} \tan^{-1} \frac{1}{3}$
21. $\frac{\pi}{3}$ 22. $\frac{\theta}{\sin \theta}$ 23. $-\frac{4}{9}$ 24. $\frac{\pi}{2}$ 25. $2(\sqrt{2}+1)$
26. $\sqrt{2}$ 27. $\frac{1}{3} \frac{a^3 - b^3}{a^2 - b^2}$ 28. $\frac{\pi}{6}$ 29. $\frac{\sqrt{3}}{2} - 1 + \frac{\pi}{6}$
30. $\frac{1}{2} \ln(2 + \sqrt{3})$ 31. $\frac{1}{4} \ln \frac{32}{17}$ 32. $\frac{\alpha}{2 \sin \alpha}$ if $\alpha \neq 0$; $\frac{1}{2}$ if $\alpha = 0$
33. 1 34. $\frac{1}{2} \ln 3$ 35. $\frac{1}{20} \ln 3$ 36. $\frac{2}{1+e}$ 37. $\ln 2$
38. 3 39. $\frac{1}{2}$ 40. $\pi/3$ 41. $1 - \frac{3}{2\sqrt{2}}$ 42. $\left(\pi - \frac{\pi^2}{4}\right)$
43. $2 - \frac{\pi}{2}(a-3b)$ 44. 13 45. 2009

Do yourself - 2

1. 2 2. $\pi \ln 2$ 3. $4\sqrt{2} - 4 \ln(\sqrt{2}+1)$ 4. $\frac{\pi}{2\sqrt{2}} - \frac{16\sqrt{2}}{5}$
5. $\frac{\pi+4}{666}$ 6. (B)

Do yourself - 3

1. 2 2. $\pi/12$ 3. $\frac{3\pi}{2}$ 4. (A) 5. (D)
7. $\frac{\pi \ln 3}{2}$ 8. (C) 9. (C) 10. (C) 11. (A)
12. (B) 13. $\frac{(a\pi+2b)\pi}{3\sqrt{3}}$ 14. $\frac{\pi(\pi+3)}{2}$ 15. (A) 16. (D)

Do yourself - 4

1. $\frac{\pi}{3}$ 2. $-\left(\frac{3\pi}{2}\right)\ln 2$ 3. 0 4. $\frac{4}{3}$ 5. (B)

Do yourself - 5

1. $\frac{23}{4}$ 2. $(\sqrt{3}-1)$ 3. (A) 4. (C) 5. (A)
6. (B) 7. (C) 8. 6 9. (C) 10. (D)

Do yourself - 6

1. 3 2. $\frac{3}{2}\sin 1$ 3. $\frac{dy}{dx} = \frac{-\sqrt{3-\sin^2 x}}{\cos y}$ 4. (A)
5. (D) 6. (B) 7. (C) 8. $a = 4$ and $b = 1$
9. (C) 10. (C) 11. (A) 12. (C)

Do yourself - 7

1. $\frac{1}{2}\ln 3$ 2. $\frac{\pi}{2}$ 3. (C) 4. (B) 5. (A)
6. (C) 7. (B) 8. 3

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	B	C	B	B	B	D	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	A	D	D	C	D	A	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	B	D	B	A	D	A	A	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C,D	A,C,D	B,C,D	A,B,C,D	A,B,C	B,C,D	A,C	B,C,D	B,C,D	C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,C,D	A,B,C,D	B,C	C,D	A,C,D	A,B,C,D	A,B	A,B,C,D	A,B	A,C,D
Que.	21	22	23	24	25					
Ans.	A,D	A	C,D	A,B,D	A,B,C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	B	A	A	B,C,D	A,B,D	4.50	5.00	6.00	A	
Q.10	A	B	C	D						
	S	P	Q	R						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	0.66 or 0.67	0.66 or 0.67	90.00	8.00	5250.00	125	153	21	5	9

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	4	1	3	3	4	2	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1 or 3 (Ans. 1 by NTA)	1	4	2	1	3	2	C	C
Que.	21	22								
Ans.	21	4								

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A,C	2	A	B	A	D	A,C,D	D	0
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	9	7	A,C	A,B	D	A,B,C	C,D	A	1	B,C
Que.	21	22	23			24	25	26	27	28
Ans.	2	B,C	Bonus (Reason : $g'(\pi/2) = g'(-\pi/2) = 0$)			B,D	1	2	1,2,3	4.00
Que.	29	30	31	32	33	34	35	36	37	38
Ans.	1,2	0.50	A,B,D	1080.00	A,B,C	2.00	1.50	C	D	182
Que.	39	40	41							
Ans.	C,D	5	B							